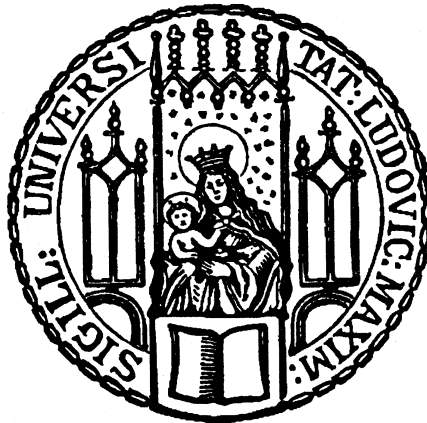

Measuring Beam Deflections via Weak Amplification in a Sagnac Interferometer

Tristan Zerweck



Munich, 5th of July, 2022

Messung von Strahlablenkungen mittels schwacher Wechselwirkung im Sagnac Interferometer

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Abstract

In this thesis, two theoretical methods will be explained, that use weak amplification in order to measure beam deflections inside and outside of an interferometer. Evaluating the altering interference pattern while performing a phase scan allow to determine the beam's spatial shift and angular displacement.

Measuring beam displacements that happen outside the interferometer is a new and unique interferometric measurement method that shall be put to the test. Therefore, a Sagnac interferometer has to be set up and aligned via the alignment method presented in this work. Afterwards, a Dove prism will be inserted into one arm of the interferometer, enabling the measurement of beam deflections outside the interferometer.

Abstract

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1 Introduction

Interferometers are frequently applied in various fields of science and technology, such as astronomy, optics, metrology and telecommunications and achieved astonishing breakthroughs in physics. For example, the Michelson-Morley experiment, an attempt to prove the existence of the aether [1], and the LIGO (Laser Interferometer Gravitational-Wave Observatory), which made the first observation of gravitational waves [2], led to the beginning of and verified special relativity. Compared to other measurement methods, distances up to a tiny length of $\frac{\lambda}{100}$ can be resolved by the interference patterns observed [3]. Optical interferometry can be used in biology, medicine and the industry to produce high resolution measurements of biomolecules, cells and material surfaces [4].

In this paper, an interferometric setup will be explained that measures beam deflections outside of an interferometer. The aim is to lay the foundation for a precise measuring instrument for just as many fruitful applications.

Therefore at first, an alignment method for interferometers will be presented [5], since these measurements can only be performed if the interferometer is well adjusted. Afterwards, an interferometric beam deflection measurement method will be explained [6].

The focus of this thesis, however, is on the experimental setup. In order to measure beam deflections outside the interferometer, both methods shall be put into practice. A Sagnac interferometer will be built and finely tuned using the alignment method. Through a phase scan, displacement parameters will be obtained, which will be used to slightly tilt one optical component inside the interferometer. After various iterations of this procedure, the Sagnac interferometer will be greatly aligned. Thereafter, the setup can be modified to put the other measurement method into practice, i.e., measuring beam displacements outside the interferometer. Hence, a dove prism will be inserted into one arm of the interferometer, after which it will again be lined up correctly. Once this is done, another phase scan will yield the displacement parameters of the outside beam deflection.

The weak interaction is crucial to the measurement as the weak amplification effect scales the beam deflection. Thus, the slightest angular displacements and transverse shifts of the laser beam are detectable.

1 Introduction

Measuring beam displacements outside an interferometer was not realized until now and is unique compared to what interferometers normally measure – beam deflections inside the interferometer. Here, however, the interference pattern is generated by the beam deflection created before the interferometer. Inside the interferometer, one beam passes through the optical components normally, while the other transits a Dove prism, which mirrors its deflection. Thus, both arms are relatively shifted and produce an interference pattern used to determine the deflection parameters.

2 Weak Measurements and a Possible Realization

In quantum mechanics, a measurement of an observable requires the wavefunction to collapse onto the observables' eigenbasis, or in other terms, every measurement is a projection onto an eigenvector. The result of this measurement lies between the largest and smallest eigenvalues. This, at least, is true for strong measurements.

In 1988 Aharonov, Albert and Vaidman (AAV) presented another concept which shows that in fact a measurement result could lie far beyond the eigenvalues of an operator if two operators only interacted weakly - the wavefunction, therefore, does **not** collapse into an eigenstate of the operator. In their original paper [7] they showed that the measurement of a spin- $\frac{1}{2}$ particle could not only yield the value $\frac{1}{2}$, but also $\langle\sigma_z\rangle = 100$ due to weak amplification.

Generally, a weak measurement consists of three steps: First, preparing the quantum state (preselection). Secondly, creating a weak interaction (e.g., with the measurement device). Thirdly, postselecting the quantum state - usually a projection onto a certain basis.

Weak measurements that produced values beyond the smallest and largest eigenvalue of an observable have been successfully carried out in quantum optics [8, 9]. The result of weak measurements can also be seen as an amplification of 'traditional' expectation values of strong measurements (hence *weak amplification*) and thus allow us to measure even smaller quantities (cf. precision measurement techniques [5, 10, 11]). This is precisely the basis of the experiment described later.

Nevertheless, one should not disregard the disadvantages of weak measurements, namely the loss of information in the postselection procedure. Clearly expressed, weak measurements trade-off amplification for information loss [10].

In addition, depending on the postselection, weak measurements are only feasible if a large quantity of particles is used because the postselection can potentially reduce the signal strongly. For these reasons, weak measurements have been claimed to be rather unfavourable [12].

2.1 Weak Values

The concept of weak values follows from the two-state vector formalism, a time-symmetric description of quantum mechanics that considers two quantum states: one in the past and one in the future, the first of which moves forward in time and the latter going back in time. Together they fully describe a quantum state, i.e., past and future measurements characterize a quantum state as a whole [13].

Consider some system and a pointer described by the observables S and P . In a quantum mechanical description an observable can be written as a Hermitian operator $\hat{A}$ ($\hat{A} = \hat{A}^\dagger$). To describe the interaction between system and pointer, the interaction operator will be defined as the tensor product $\hat{A} = \hat{S} \otimes \hat{P}$.

A Hermitian operator can generate a continuous transformation via the unitary operator $\hat{U}(\epsilon) = \exp(-i\epsilon\hat{A})$ along ϵ . For example, if $\hat{A}$ is a Hamiltonian, then the unitary operator $\hat{U}(\epsilon) = \exp(-i\epsilon\hat{A})$ describes a translation in time by ϵ , analogously $\hat{P}$, the momentum operator, generates a translation in space, etc [14].

Furthermore, ϵ quantifies the coupling strength between the system and the pointer. Now let us set ϵ to be very small, i.e. let us make the interaction between system and pointer very small (*weak* coupling).

In this way, the unitary operator can be rewritten as it's Taylor expansion which approximates sufficiently well for small ϵ .

$$\hat{U}(\epsilon) = \exp(-i\epsilon\hat{A}) = \mathbb{1} - i\epsilon\hat{A} - \frac{\epsilon^2}{2}\hat{A}^2 + O(\epsilon^3). \quad (2.1)$$

In quantum mechanics the probability of measuring an unperturbed system in state $|\psi'\rangle$ after it being in state $|\psi\rangle$ is given by $P = |\langle\psi|\psi'\rangle|^2$ while $|\psi\rangle$ and $|\psi'\rangle$ characterize the preselected and postselected state of a system.

When the initial state has been modified by some interaction operator $\hat{U}(\epsilon)$, the probability P also differs. It can be expressed by:

$$P_\epsilon = |\langle\psi|\hat{U}(\epsilon)|\psi'\rangle|^2 = |\langle\psi|\mathbb{1} - i\epsilon\hat{A} - \frac{\epsilon^2}{2}\hat{A}^2 + O(\epsilon^3)|\psi'\rangle|^2, \quad (2.2)$$

$$= P - |\langle\psi|i\epsilon\hat{A}|\psi'\rangle|^2 + O(\epsilon^3). \quad (2.3)$$

Therefore the relative correction $\frac{P_\epsilon}{P}$ can then be written as

$$\frac{P_\epsilon}{P} = 1 + 2\epsilon \text{Im}(A_w) - \epsilon^2[\text{Re } A_w^2 - |A_w|^2] + O(\epsilon^3), \quad (2.4)$$

while the short notation A_w was used to describe the weak value of first order of an observable operator $\hat{A}$ for pre- and postselected state of a system $|\psi\rangle$ and $|\psi'\rangle$ [14].

$$A_w = \frac{\langle\psi'|\hat{A}|\psi\rangle}{\langle\psi'|\psi\rangle}. \quad (2.5)$$

Analogously, weak values can also be defined for higher orders of ϵ

$$A_w^n = \frac{\langle \psi' | \hat{A}^n | \psi \rangle}{\langle \psi' | \psi \rangle}. \quad (2.6)$$

Nevertheless, most weak measurement experiments are conducted in the weak interaction regime where the probability correction is linearly proportional to the weak value and higher orders of ϵ can be neglected [14–17]. This will also be the case for the interferometer described in this paper.

In summary, the expectation value of a perturbed system, described by the operator $\hat{U}(\epsilon) = \exp(-i\epsilon\hat{A})$, is changed by some value proportional to the weak value, which can potentially result in great amplifications, like suggested in the title of the original weak value paper (*How the result of a measurement of a component of the spin of a spin-1/2 particle can turn out to be 100*) [7].

2.2 Determining Weak Values

In the following pages, a general mathematical approach will be put forth, describing the measurement of weak values. More precisely, a general scheme will be explained of describing measurement results of observables in weak measurements, similar to expectation values, which are characterized by weak values. This can generally be used for every type of weak measurement and was originally presented by AAV in [7]. However, I will follow a notation similar to the one presented in [6].

Imagine a system characterized by the wave function $|\psi_S\rangle$ and a pointer defined by the state $|\psi_P\rangle$. Now let us express a weak interaction as $U = e^{-i\epsilon\hat{A}\otimes\hat{B}}$ with ϵ the coupling strength and an operator $\hat{A}$, which is an observable of the system, and a second operator $\hat{B}$, which represents an operator acting on the pointer. The coupled state can then be written as

$$\hat{U}|\psi_S\rangle \otimes |\psi_P\rangle = e^{-i\epsilon\hat{A}\otimes\hat{B}}|\psi_S\rangle \otimes |\psi_P\rangle. \quad (2.7)$$

The coupled state is then projected onto the final state of the system $|\psi'_S\rangle$. This is the postselection which can be represented by

$$|\psi'_P\rangle = \mathcal{N}\langle\psi'_S| \left(e^{-i\epsilon\hat{A}\otimes\hat{B}}|\psi_S\rangle \otimes |\psi_P\rangle \right), \quad (2.8)$$

resulting in the final state of the pointer, marked by the apostrophe in $|\psi'_P\rangle$. The normalisation factor is described by $\mathcal{N}$.

2 Weak Measurements and a Possible Realization

The term in the previous Equation (2.8) can be simplified by utilizing that ϵ is very small and therefore approximating the exponential function. Additionally, the shorthand notation $|\psi_S\rangle|\psi_P\rangle$ for the tensor product $|\psi_S\rangle \otimes |\psi_P\rangle$ was adopted.

$$\begin{aligned}
|\psi'_P\rangle &= \mathcal{N}\langle\psi'_S| \left(e^{-i\epsilon\hat{A}\otimes\hat{B}} |\psi_S\rangle \otimes |\psi_P\rangle \right) \\
&\approx \mathcal{N}\langle\psi'_S| (\mathbb{1} - i\epsilon\hat{A} \otimes \hat{B}) |\psi_S\rangle |\psi_P\rangle \\
&= \mathcal{N}\langle\psi'_S | \psi_S\rangle |\psi_P\rangle - i\epsilon\langle\psi'_S|\hat{A}|\psi_S\rangle\hat{B}|\psi_S\rangle|\psi_P\rangle \\
&= \mathcal{N}\left(\mathbb{1} - i\epsilon \frac{\langle\psi'_S|\hat{A}|\psi_S\rangle}{\langle\psi'_S|\psi_S\rangle} \hat{B} \right) |\psi_P\rangle \\
&= \mathcal{N}\left(\mathbb{1} - i\epsilon A_w \hat{B} \right) |\psi_P\rangle.
\end{aligned} \tag{2.9}$$

Here the definition of the weak value A_w from Equation (2.5) comes into play again.

What is more, an expectation value $\langle\hat{O}\rangle$ of a pointer observable $\hat{O}$ can now be written as

$$\begin{aligned}
\langle\hat{O}\rangle_{P'} &= \langle\psi'_P|\hat{O}|\psi'_P\rangle \\
&\approx \langle\psi_P| \left(\mathbb{1} + i\epsilon A_w^* \hat{B} \right) \hat{O} \left(\mathbb{1} - i\epsilon A_w \hat{B} \right) |\psi_P\rangle \\
&= \langle\psi_P|\hat{O}|\psi_P\rangle + i\epsilon A_w^* \langle\psi_P|\hat{B}\hat{O}|\psi_P\rangle - i\epsilon A_w \langle\psi_P|\hat{O}\hat{B}|\psi_P\rangle \\
&= \langle\psi_P|\hat{O}|\psi_P\rangle + i\epsilon \text{Re}(A_w) \langle\psi_P| [\hat{B}, \hat{O}] |\psi_P\rangle \\
&\quad - \epsilon \text{Im}(A_w) \langle\psi_P| \{ \hat{B}, \hat{O} \} |\psi_P\rangle.
\end{aligned} \tag{2.10}$$

In this way, the measurable pointer shift can be defined as the difference between the expectation value of the non-displaced pointer and the disturbed pointer state. In a special case the following result can be obtained;

$$\delta O = \langle\psi'_P|\hat{O}|\psi'_P\rangle - \langle\psi_P|\hat{O}|\psi_P\rangle = \epsilon \text{Re}(A_w). \tag{2.11}$$

In this situation, the operators $\hat{B}$ and $\hat{O}$ must be canonical conjugate quantities.¹ Therefore the commutator $[\hat{B}, \hat{O}] = i$ (with $\hbar = 1$). Additionally, $\langle\psi_P| \{ \hat{B}, \hat{O} \} |\psi_P\rangle = 0$, the anticommutators vanish, because of an experimental condition, explained in section 3.6.

¹One quantity is the Fourier transform of the other, e.g., $\hat{x}$ and $\hat{p}$.

Analogously, the expectation value of the operator $\hat{B}$, which acts on the pointer state, can be written as

$$\begin{aligned}
 \langle \hat{B} \rangle_{P'} &= \langle \psi'_P | \hat{B} | \psi'_P \rangle \\
 &\approx \langle \psi_P | \left(\mathbb{1} + i\epsilon A_w^* \hat{B} \right) \hat{B} \left(\mathbb{1} - i\epsilon A_w \hat{B} \right) | \psi_P \rangle \\
 &= \langle \psi_P | \hat{B} | \psi_P \rangle + i\epsilon A_w^* \langle \psi_P | \hat{B}^2 | \psi_P \rangle - i\epsilon A_w \langle \psi_P | \hat{B}^2 | \psi_P \rangle \\
 &= \langle \psi_P | \hat{B} | \psi_P \rangle + 2\epsilon \text{Im}(A_w) \langle \psi_P | \hat{B}^2 | \psi_P \rangle \\
 &= \langle \psi_P | \hat{B} | \psi_P \rangle + 2\epsilon \text{Im}(A_w) \Delta B^2.
 \end{aligned} \tag{2.12}$$

The shift of the expectation values is again simplified in the special case to yield the following formula

$$\delta B = \langle \psi'_P | \hat{B} | \psi'_P \rangle - \langle \psi_P | \hat{B} | \psi_P \rangle = 2\epsilon \text{Im}(A_w) \Delta B^2. \tag{2.13}$$

In the last transformations the variance ΔB was introduced. The variance of $\hat{B}$ is given by

$$\Delta B^2 = \langle \psi_P | \hat{B}^2 | \psi_P \rangle - \underbrace{\langle \psi_P | \hat{B} | \psi_P \rangle^2}_{=0}, \tag{2.14}$$

where it was used that the pointer operator does not act on the unperturbed pointer state, therefore $\langle \psi_P | \hat{B} | \psi_P \rangle = 0$.

From this result, one can again clearly see the dependence of the pointer expectation value or the measured shift on the weak value, reflecting the power of weak measurements. The experimenter can choose the weak value so that conventional expectation values in strong measurements are amplified tremendously through the factor of the weak value. Choosing the weak value A_w happens through fixing the pre-and postselected state $|\psi_S\rangle$ and $|\psi'_S\rangle$. Thus more precise measurements than standard quantum mechanical measurements can be undertaken, as shown in the experiment.

2.3 Mach-Zehnder Interferometer

The theory described in the previous section is now applied to a measurement in a Mach-Zehnder interferometer (MZI). The Mach-Zehnder interferometer is one of the simplest types of interferometers. The concepts presented in the following pages can be transferred efficiently to other interferometers, like the Sagnac interferometer. In Figure 2.1 a MZI is depicted. The first beam splitter (BS-1) divides the incoming beam (BS-1); half of the beam is transmitted, and the other half is reflected. Then both beams are redirected by mirrors (M-1/2), at a 45° angle, to meet at the second beam splitter (BS-2), where again half of each beam is reflected or transmitted and an interference pattern is created. Instead of a laser beam, single photons could be used in a superposition of arms A and B. However, the results would be the same since probability distributions of photons correspond to beam intensity distributions.

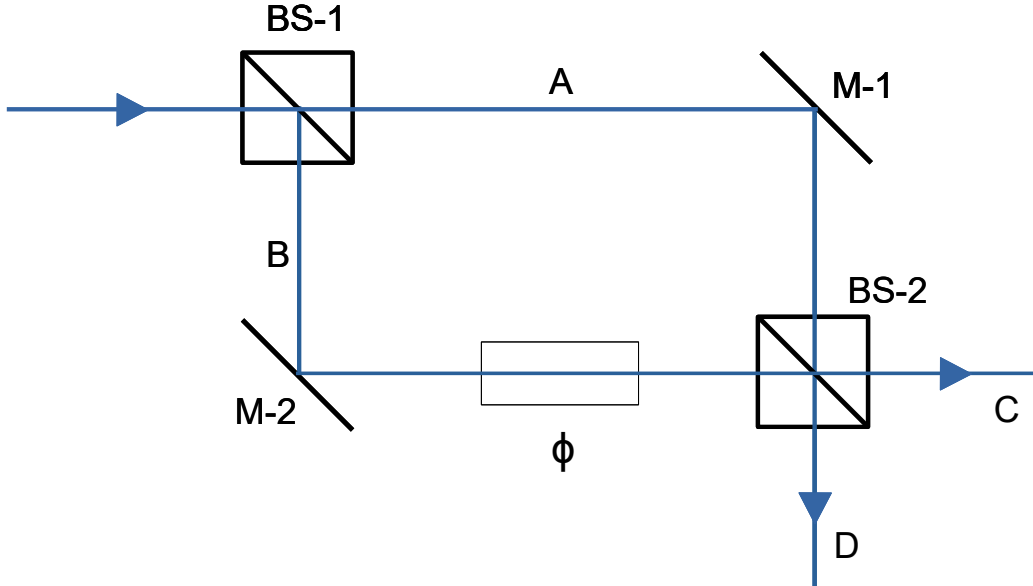


Figure 2.1: General Structure of a Mach-Zehnder-Interferometer.

Reflections at beam splitters shift the phase of the light by $\frac{\pi}{2}$. If both arms A and B are equally long, then the beam being reflected by both beam splitters (on path B and D) is shifted by π in total while the other beam passing through both beam splitters (taking path A and C) is not shifted at all. Therefore one will observe destructive interference at port D and, respectively, constructive interference in arm C [18, 19].

Nonetheless, if both arms A and B do not have the same length, which is the general case, the phase shift must also be considered. Furthermore, one beam

2.3 Mach-Zehnder Interferometer

passing through a different medium can also introduce a phase shift. Altogether, the global phase shift is represented in Figure 2.1 by the box marked as ϕ . The intensities at both output ports C and D are then dependent on the interference of both beams and therefore dependent on the phase shift ϕ .

In quantum mechanics, the system $|\Psi\rangle$ inside the interferometer can be expressed as a superposition of eigenstates of both arms $|A\rangle$ and $|B\rangle$, characterizing the spatial position of the system - later in the experiment, this system will be represented by the superposition state of photons in arm A and B. Together with the relative phase shift ϕ in one arm $|\Psi\rangle$ can be expressed as:

$$|\Psi\rangle = |A\rangle + e^{i\phi}|B\rangle. \quad (2.15)$$

Furthermore, the other path eigenstates $|C\rangle$ and $|D\rangle$ are equal to

$$|C\rangle = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle), \quad (2.16)$$

$$|D\rangle = \frac{1}{\sqrt{2}}(|A\rangle - |B\rangle), \quad (2.17)$$

which represent the constructive and destructive interference in path C and D [5],[6]. From this, it directly follows, that the intensities at the output ports C and D are given by

$$I_C = |\langle C | \Psi \rangle|^2 = 2 \sin^2\left(\frac{\phi}{2}\right), \quad (2.18)$$

$$I_D = |\langle D | \Psi \rangle|^2 = 2 \cos^2\left(\frac{\phi}{2}\right). \quad (2.19)$$

Lastly, an essential characteristic of every interferometer has to be explained. The *visibility* $\mathcal{V}$ is defined via

$$\mathcal{V} = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}. \quad (2.20)$$

It lies in the interval $[0,1]$, 0 being the worst visibility ($I_{max} = I_{min}$) and 1 being the best (i.e. $I_{min} = 0$) [20].

Various adaptations of this will be used later to characterize the interferometer.

2.3.1 Misaligned Mach-Zehnder Interferometer

Aligning a Mach-Zehnder interferometer is a complicated and challenging undertaking. Tiny imprecisions in the build-up can spoil interference patterns. For example, how does the interference change if one mirror is slightly tilted? First, if no mirror is not tilted, the interference pattern remains untouched. Even when the incoming beam enters the interferometer with a slight angular and spatial displacement, this would not be problematic as long as both mirrors were perfectly aligned at a 45° degree angle. Hence, the split beams would still overlap at the same place at the second beam splitter because both beams experienced the same displacement. Therefore, they would produce the same interference pattern as without the beam shift.

But what if one of the mirrors is not aligned perfectly, i.e., not aligned correctly to the extent of our measuring capabilities?

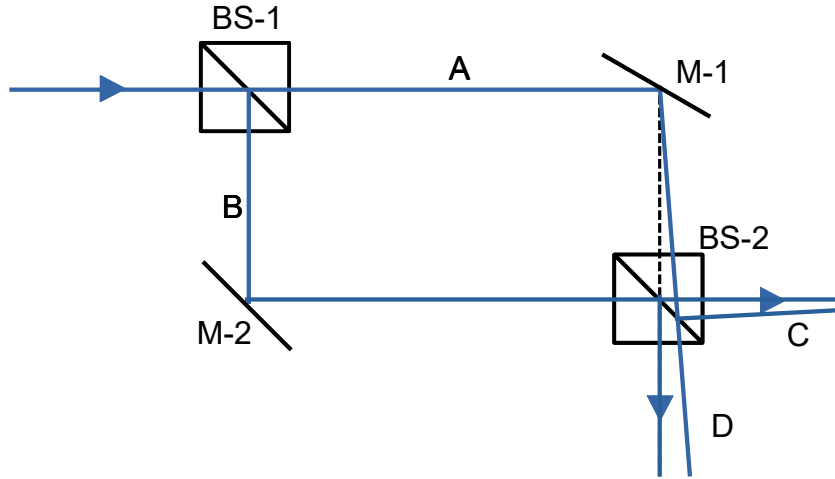


Figure 2.2: MZI with Tilted Mirror.

Figure 2.2 illustrates a single tilted mirror and the resulting beam's path. Both arms are equally long, and the phase shift ϕ vanishes. Therefore, constructive interference is measured at port C (bright port) and destructive interference at port D (dark port), as explained in the previous section. Only port C, with constructively interfering beams, is monitored. When the outgoing beams in port C (see Figure 2.2) are not parallel but diverge at an angle of θ , the wavefronts overlap and interfere. In this case, interference fringes will be observed instead of a pure gaussian intensity distribution.

Figure 2.3 illustrates two beams with plane wavefronts interfering as a result of an angle θ between the directions of beam propagation. The bold black parallel lines in the picture represent the wave fronts of the beams. The overlap generates the interference fringes. An analogon many are familiar with are the interfering

2.3 Mach-Zehnder Interferometer

wavefronts in the double-slit experiment. The screen on the right, only illustrated by a thin line, measures the intensity distribution which is proportional to the $\text{sinc}(x)^2$ function ($\text{sinc}(x) = \frac{\sin(x)}{x}$) which is depicted vertically on the far right.

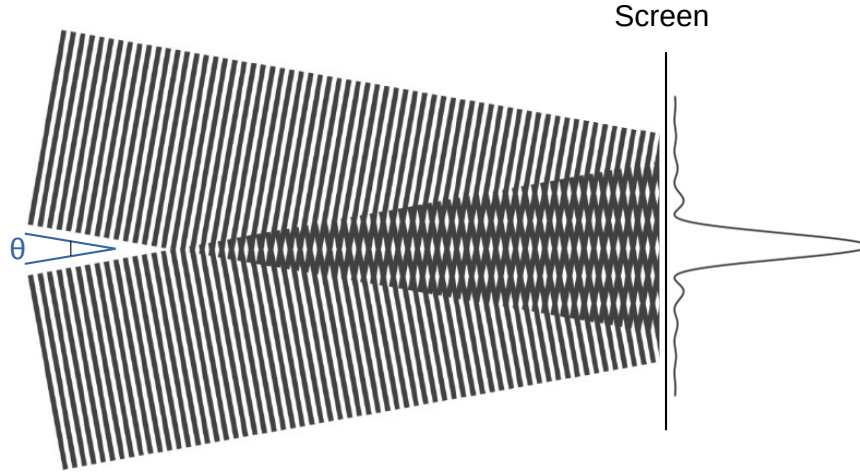


Figure 2.3: Two Beams Interfering at an Angle θ . Modified from [21].

In a better aligned MZI, the angle θ between both beams disappears. Then there are no interference fringes, but solely an overlap of two Gaussian functions since the intensity distribution of the laser beams are Gaussians. In Figure 2.4 this has been visualized: the intensity measured by the detector (at port C) is found on the right in black colour. The black curve is the sum of the blue and green Gaussian functions, representing the intensity distribution of each arm's beams. In a perfectly aligned MZI, both blue and green curves overlap, so the black curve would be minimally narrow and not as stretched out as in the figure. Note that the sum of both Gaussians can only be computed because there is no phase shift between both arms in this case. If there were a phase shift, the Gaussians would interfere destructively. Nevertheless, no interference fringes would be generated because the beams are parallel [21].

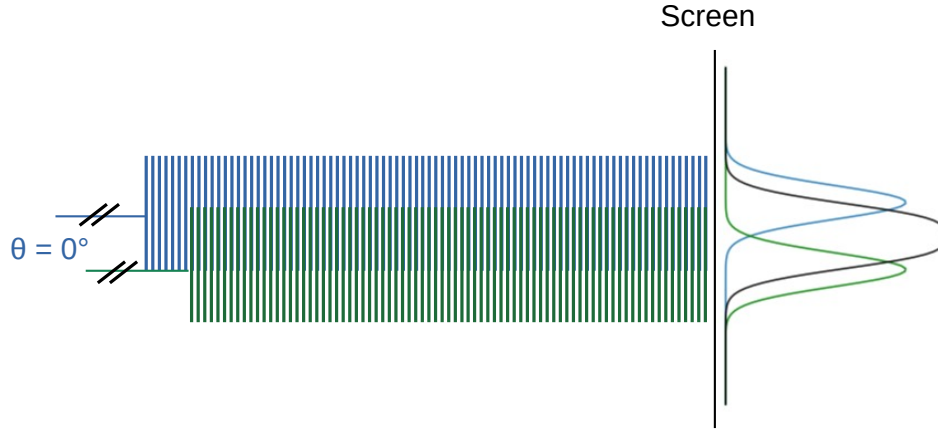


Figure 2.4: Two Parallel Beams Overlapping.

How can one align an interferometer by correcting the tilted mirror and, in this way, reducing the angle θ between the beams?

In this paper, [5] Dziewior et al., among other things, presented an interferometric alignment method, which explains how an interferometer can be iteratively aligned via a phase scan. Thus, one can determine a beam's angular and spatial displacement (δx and $\delta \theta$) and set the mirrors at the right angle. This alignment method will be used in the experimental part to measure δx and $\delta \theta$ and to line up the interferometer.

The following pages comprising the alignment method strongly follow the notation presented in [5]. It can be seen as building on the general weak measurement theory of section 2.2.

Firstly, the state of the system $|\psi_S\rangle$ in the MZI with a tilted mirror (Figure 2.2) is expressed by

$$|\psi_S\rangle = \cos(\alpha)|A\rangle + \sin(\alpha)e^{i\phi}|B\rangle. \quad (2.21)$$

This formula simply describes the system being in a superposition state of arms A and B, taking into account the phase shift ϕ due to the path difference in the MZIs arms. In section 2.3 a very similar formula was already deduced, the difference being here that the $\cos(\alpha)$ and $\sin(\alpha)$ terms are now present because the intensities in both arms vary mutually. With this formula, states can be characterized where the intensity of the initial beam is not split 50:50 for both arms but in another ratio.

2.3 Mach-Zehnder Interferometer

Additionally, the intensities at each output port (C or D) for this state are expressed by

$$\begin{aligned} I_C &= |\langle C | \psi_S \rangle|^2, \\ I_D &= |\langle D | \psi_S \rangle|^2, \\ I_{C/D} &= \frac{1}{2} \pm \cos(\alpha) \sin(\alpha) \cos(\phi), \end{aligned} \tag{2.22}$$

where again the projection onto arm C or D from the previous section was used (cf. Equation (2.16)). The maximum intensity is found at port C (constructive interference) and the minimum intensity at port D (destructive interference) as written earlier.

Therefore the visibility (cf. Equation (2.20)) can also be characterized by

$$\mathcal{V} = 2 \cos(\alpha) \sin(\alpha) = \sin(2\alpha). \tag{2.23}$$

Secondly, a description of the pointer state has to be given in order to characterize the coupled state $|\psi\rangle$. The pointer is realized by a degree of freedom of the photon itself – the spatial mode. It can either be in the initial non-perturbed state $|\chi\rangle$ or in the shifted state $|\chi'\rangle$ - the shifted state $|\chi'\rangle$ is coupled to the photons in arm B because the second mirror is tilted, as shown in Figure 2.2.

The perturbed state is more precisely characterized as

$$|\chi'\rangle = \eta (|\chi\rangle + \epsilon |\chi^\perp\rangle). \tag{2.24}$$

Here $|\chi^\perp\rangle$ denotes a tiny displacement of the initial spatial mode in the direction of an orthogonal mode, where the scalarproduct $\langle \chi | \chi^\perp \rangle$ vanishes. While η defines the overlap and normalization $\eta = \langle \chi | \chi' \rangle = \frac{1}{\sqrt{1+\epsilon^2}}$.

Following this, the coupled state can be written as

$$|\psi\rangle = \cos(\alpha) |A\rangle |\chi'\rangle + \sin(\alpha) e^{i\phi} |B\rangle |\chi\rangle. \tag{2.25}$$

The pointer is being shifted when photons are in arm A, and the pointer is not being shifted in arm B. Furthermore, there is a relative phase shift between both arms, expressed by $e^{i\phi}$, and an unequal intensity distribution in both arms characterized by the cosine and sine functions. In the rest of the paper, the ratio between the intensities will often be described by $\frac{\sin(\alpha)}{\cos(\alpha)} = \tan(\alpha)$.

Corresponding to Equation (2.23), the visibility can now be written as

$$\mathcal{V} = \eta \sin(2\alpha), \tag{2.26}$$

where the intensities were computed analogously as presented above in Equation (2.22). This relation between η , the visibility $\mathcal{V}$ and α will be essential later.

2 Weak Measurements and a Possible Realization

Now the postselection process will be represented by the projection onto arm C, using $|C\rangle = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle)$. The final pointer state is then given by

$$|\psi'_P\rangle = \mathcal{N} \left(|\chi\rangle + \frac{\eta\epsilon}{\eta + \tan(\alpha)e^{i\phi}} |\chi^\perp\rangle \right). \quad (2.27)$$

Since the weak value of the projection on A (cf. Equation (2.5)) is characterized by

$$(\mathbf{P}_A)_w = (|A\rangle\langle A|)_w = \frac{\langle C | A \rangle \langle A | \psi \rangle}{\langle C | \psi \rangle} = \frac{1}{1 + \tan(\alpha)e^{i\phi}}, \quad (2.28)$$

the final pointer state, as expected from the previous section, can be rewritten as

$$|\psi'_P\rangle = |\chi\rangle + \epsilon(\mathbf{P}_A)_w |\chi^\perp\rangle. \quad (2.29)$$

Here $\epsilon \ll 1$ was used, wherefore η becomes 1. This is only the case for very small tilts of the mirror. Note again, that the shift of the pointer state is linearly proportional to the weak value $(\mathbf{P}_A)_w$.

Nevertheless, in the more general case, where a finite coupling strength and imperfections of the setup have to be considered, a mixed state $|\psi\rangle$ has to be characterized by the density matrix

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix} \cos(\alpha)^2 & \cos(\alpha)\sin(\alpha)e^{-i\phi}\eta \\ \cos(\alpha)\sin(\alpha)e^{i\phi}\eta & \sin(\alpha)^2 \end{pmatrix}. \quad (2.30)$$

In the case of mixed states, the weak value can be calculated by

$$A_w = \frac{\text{Tr}(\rho_{\text{post}} A \rho_{\text{pre}})}{\text{Tr}(\rho_{\text{post}} \rho_{\text{pre}})}, \quad (2.31)$$

from which follows that the weak value is given by

$$(\mathbf{P}_A)_w = \frac{\text{Tr}(|C\rangle\langle C|(\mathbf{P}_A)_w \rho)}{\text{Tr}(|C\rangle\langle C|\rho)} = \frac{1 + \tan(\alpha)\eta e^{-i\phi}}{1 + \tan(\alpha)^2 + 2 \tan(\alpha)\eta \cos(\phi)}. \quad (2.32)$$

Interactions and imperfections reduce the overlap η . For $\eta = 1$ the weak value is maximal; for smaller values of η the weak value decreases. Therefore, the greatest possible η has to be achieved experimentally. This can be realized by maximizing the visibility and adjusting α in Equation (2.26) [5].

Correspondingly to Equation (2.11), the change of an expectation value of an observable $\hat{O}$ from the non-perturbed to the shifted state could in a special case (where $[\hat{B}, \hat{O}] = \{\hat{B}, \hat{O}\} = 0$) be interpreted as:

$$\delta O = \langle \chi' | \hat{O} | \chi' \rangle - \langle \chi | \hat{O} | \chi \rangle = 2\epsilon \text{Re}[\langle \chi | \hat{O} | \chi^\perp \rangle (\mathbf{P}_A)_w]. \quad (2.33)$$

These formulae are relevant to the experiment once the wavefunctions of the pointer state are defined to calculate the expectation values.

The pointer is a photon beam in the z-direction with a Gaussian intensity distribution in both x and y directions. The unperturbed and perturbed pointer wavefunction can be written as

$$\chi(Q) = e^{-Q^2/4(\Delta Q)^2}, \quad (2.34)$$

$$\chi'(Q) = \chi(Q - \delta Q) = e^{-(Q-\delta Q)^2/4(\Delta(Q-\delta Q))^2}, \quad (2.35)$$

with a general pointer variable Q . This Q will later be replaced by the position or momentum variables. The interaction in the second arm (with the tilted mirror) displaces Q by δQ . This also shifts the expectation value $\delta\langle Q \rangle = \delta Q$. The expressions for the shifts of the expectation values δO stay the same with the Gaussian pointer as before in Equation (2.33). Moreover, $\langle \chi|Q|\chi \rangle = 0$ and $\langle \chi'|Q|\chi' \rangle = \delta Q$. Likewise, the shift of the expectation value in momentum space δP_Q can be expressed similarly since the Gaussian in the position variable Q is also a Gaussian in its conjugate momentum representation.

Consider now the operators $\hat{O}$ and $\hat{B}$ to be specific operators. Let $\hat{B} = \epsilon_x \hat{x} + \epsilon_k \hat{k}$, the operator acting on the pointer, represent the beam deflection caused by the momentum operator $\hat{k}$ and the angular displacement generated by the position operator $\hat{x}$. Furthermore, the measured pointer observable $\hat{O}$ can either be the position $\hat{x}$ or the momentum $\hat{k}$.

Then the shift of the expectation value in position or momentum space can be written as

$$\delta x = 2\epsilon \text{Re}[\langle \chi|\hat{x}|\chi^\perp \rangle] \text{Re}(\mathbf{P}_A)_w - 4\epsilon \text{Re}[\langle \chi|\hat{k}|\chi^\perp \rangle](\Delta x)^2 \text{Im}(\mathbf{P}_A)_w, \quad (2.36)$$

$$\delta k = 2\epsilon \text{Re}[\langle \chi|\hat{k}|\chi^\perp \rangle] \text{Re}(\mathbf{P}_A)_w + \frac{2\epsilon \text{Re}[\langle \chi|\hat{x}|\chi^\perp \rangle]}{2(\Delta x)^2} \text{Im}(\mathbf{P}_A)_w. \quad (2.37)$$

The sum in these equations can be understood intuitively. The measured spatial shift δx of the pointer is given by both the beam displacement (first term) and the angular shift (second term). Analogously, the sum in the measured momentum shift can be explained this way.

2 Weak Measurements and a Possible Realization

From this and $\delta p_y = \frac{2\pi}{\lambda} \delta \theta x$ it follows that the overall shift of the pointer $\delta \mathbf{R}$, i.e. the measured shift on the detector, in both x and y direction due to an angular and a spatial displacement is given by

$$\delta R_x = (\delta x + z \delta \theta y) \operatorname{Re}(\mathbf{P}_A)_w + \left(\frac{z}{z_R} \delta x - z_R \delta \theta y \right) \operatorname{Im}(\mathbf{P}_A)_w, \quad (2.38)$$

$$\delta R_y = (\delta y - z \delta \theta x) \operatorname{Re}(\mathbf{P}_A)_w + \left(\frac{z}{z_R} \delta y + z_R \delta \theta x \right) \operatorname{Im}(\mathbf{P}_A)_w. \quad (2.39)$$

Here the Rayleigh length $z_R = \frac{\pi w_0^2}{\lambda}$ was used and the fact that the shift on the detector $\delta \mathbf{R}$ is the sum of the positional displacement $\delta \mathbf{r}$ and the angular shift given by $\delta \boldsymbol{\theta} \times \mathbf{z}$. The vector $\mathbf{z} = (0, 0, z)$ runs along the beam, and the length z defines the distance between the measuring device and the beam's focus.

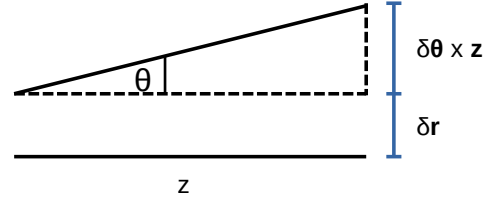


Figure 2.5: Sketch of Beam Deflection.

The mathematical derivation of Equations (2.38), (2.39) can be found more detailed in said paper [5].

Concluding, these last two equations will be used in the experimental part as an alignment method for the interferometer. Data will be fitted to the Equations (2.38) and (2.39) in order to attain the parameters for the spatial shift $\delta \mathbf{r}$ and the angle $\delta \boldsymbol{\theta}$. These two parameters will be the only unknown values. The other values that will be determined beforehand include z, z_R, w_0, α and ϕ , the last two of which are hidden in the weak value $(\mathbf{P}_A)_w$. The value z will simply be fixed by measuring the distance between the measuring device and the beam's focus. The Rayleigh length z_R will be ascertained by determining the beam width w_0 at the focus (the method for this measurement will be explained later). The α value can be detected by monitoring the intensities in both beam arms. And lastly, ϕ will be measured by shifting the phase and observing the intensity evolution.

After having worked out $\delta \mathbf{r}$ and $\delta \boldsymbol{\theta}$, the mirror will be slightly shifted, and again the spatial shift and angular displacement will be fitted. This will be repeated iteratively until the shift vanishes. The exact experimental procedure is explained in section 3.

2.3.2 Mach-Zehnder Interferometer with Dove Prism

The modified Mach-Zehnder interferometer outlined in this section continues the idea of measuring the deflection of a beam. However, the concept is extended to measuring beam deflections outside the interferometer.

Imagine a tilted beam entering a Mach-Zehnder interferometer. How could one measure how strongly the direction deviates?

As mentioned at the beginning of chapter 2.2, if the interferometer were perfectly aligned, one would not notice if the incoming beam was tilted at the output port, as both arms would be equally affected by the deflection. One would measure the same as if the beam was not tilted. Nevertheless, if a Dove prism is placed inside one arm of the interferometer, then both beams will be relatively displaced. The Dove prism deflects a beam passing through in the opposite direction by the same amount, i.e., the angle and spatial shift are flipped around the central axis. In Figure 2.6 the beam's path inside the Dove prism can be understood and how the beam is turned upside down.

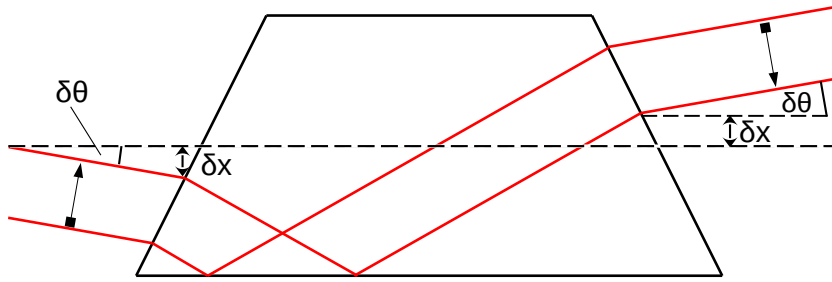


Figure 2.6: Beam Path inside Dove Prism. Modified from [6].

Therefore, instead of having a displaced mirror inside the interferometer causing the beam shift, a Dove prism will deflect the beam, to consequently make the initial shift visible through interference patterns at the output port C.

This method's great advantage is that beam deflections outside the interferometer can now be measured. In contrast, most interferometers only show the beams deflected inside them, like in [5].

The following theoretical description of the interferometer set up in this chapter fully originates from Köster's bachelor thesis [6]. Figure 2.7 schematically depicts the setup with the Dove prism and the beams unperturbed path (dotted black line) and perturbed path (blue line). It can be seen how the reflection inside the Dove prism causes interference at the output ports.

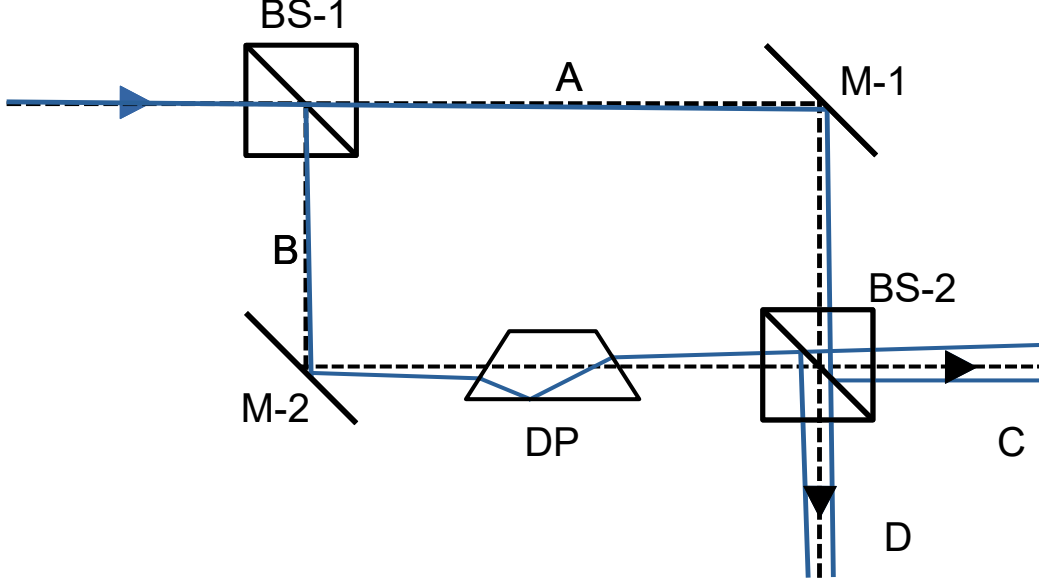


Figure 2.7: Mach-Zehnder Interferometer with Dove Prism. Adapted from [6].

Analogously, a mathematical description of this setup can be derived similarly to the general representation in section 2.2 and to what was already used in section 2.3.1.

Primo, the system's state and the coupler's state must be defined. For the system, the previous notation $|\psi_S\rangle$ will be reused.

$$|\psi_S\rangle = \cos(\alpha)|A\rangle + \sin(\alpha)e^{i\phi}|B\rangle. \quad (2.40)$$

The notation $|\chi\rangle$ or $|\chi'\rangle$ for the perturbed and unperturbed beam from the preceding section will again be taken for the pointer state $|\psi_P\rangle$.

$$|\chi'\rangle = \eta (|\chi\rangle + \epsilon|\chi^\perp\rangle). \quad (2.41)$$

With $\langle\chi|\chi^\perp\rangle = 0$. And η the normalization $\eta = \langle\chi|\chi'\rangle = \frac{1}{\sqrt{1+\epsilon^2}}$. The composite state or coupled system is then represented by

$$|\psi\rangle = \cos(\alpha)|A\rangle|\chi'\rangle + \sin(\alpha)e^{i\phi}|B\rangle\hat{D}|\chi'\rangle. \quad (2.42)$$

Notice that both arms are in the perturbed state $|\chi'\rangle$ since the beam entering the interferometer is already deflected.

Furthermore $\hat{D}$ denotes an operator characterizing the interaction with the Dove prism, which only acts on arm B, since the Dove prism is positioned there.

The operator $\hat{D}$ acts on $|\chi'\rangle$ or its components as follows

$$\begin{aligned} \hat{D}|\chi\rangle &= |\chi\rangle, \\ \hat{D}|\chi^\perp\rangle &= -|\chi^\perp\rangle. \end{aligned} \quad (2.43)$$

2.3 Mach-Zehnder Interferometer

These equations represent that the Dove prism does not change the unperturbed beam. This is also visualized in Figure 2.7 by the dotted black line going through the Dove prism without changing being deflected. Nevertheless, the interaction $\hat{D}|\chi^\perp\rangle$ of the Dove prism with the displaced beam is characterized by flipping the spatial shift and angle of the beam relative to the prism axis.

In the next step, after defining the state of the system and the pointer and thus the coupled state, the postselected state will be described. This will again be done by projecting onto $|C\rangle = \frac{1}{\sqrt{2}}(|A\rangle + |B\rangle)$, the output arm:

$$|\psi'_F\rangle = |\chi\rangle - 2\epsilon(\mathbf{P}_B)_w|\chi^\perp\rangle + \epsilon|\chi^\perp\rangle, \quad (2.44)$$

where correspondingly to Equation (2.32) the final pointer state is characterized by the projector $(\mathbf{P}_B)_w$.

The expectation value of an arbitrary operator $\hat{O}$ in the final pointer state (cf. Equation (2.10)) is therefore likewise given by

$$\begin{aligned} \langle\hat{O}\rangle_{F'} &= \langle\psi'_F|\hat{O}|\psi'_F\rangle, \\ &= \langle\chi|\hat{O}|\chi\rangle - 4\epsilon \text{Re}[(\mathbf{P}_B)_w\langle\chi|\hat{O}|\chi^\perp\rangle] + 2\epsilon \text{Re}[\langle\chi|\hat{O}|\chi^\perp\rangle]. \end{aligned} \quad (2.45)$$

Therefore, the shift of the expectation value is given by

$$\begin{aligned} \delta O &= \langle\psi'_F|\hat{O}|\psi'_F\rangle - \langle\psi_F|\hat{O}|\psi_F\rangle = \langle\psi'_F|\hat{O}|\psi'_F\rangle - \langle\chi|\hat{O}|\chi\rangle, \\ &= -4\epsilon \text{Re}[(\mathbf{P}_B)_w\langle\chi|\hat{O}|\chi^\perp\rangle] + 2\epsilon \text{Re}[\langle\chi|\hat{O}|\chi^\perp\rangle]. \end{aligned} \quad (2.46)$$

Now that the expectation value is defined, δx and $\delta\theta$, the shift in space and the angle of the beam in one dimension, have to be computed.

The interaction outside the interferometer must first be expressed. The operator $\hat{B}$ will be defined as in the previous section: $\hat{B} = \epsilon_x \hat{k} - \epsilon_k \hat{x}$. It outlines the momentum and position shift that deflects the beam outside the interferometer. When acting on the nonperturbed beam $|\chi\rangle$ the state is shifted to the perturbed state:

$$e^{-i\hat{B}}|\chi\rangle = |\chi'\rangle. \quad (2.47)$$

Therefore, the operator $\hat{D}$ of the Dove prism acting on the shifted state can be defined as

$$\hat{D}|\chi'\rangle = \hat{D}e^{-i\hat{B}}|\chi\rangle = e^{i\hat{B}}|\chi\rangle. \quad (2.48)$$

This equation reflects the property of the Dove prism of mirroring the deflection. Nevertheless, for finding the expectation value's shift of δx as in Equation (2.46) one needs to compute how $\hat{x}$ acts on $|\chi^\perp\rangle$. For that reason we will rewrite $|\chi^\perp\rangle$, by

2 Weak Measurements and a Possible Realization

using Equation (2.41) first, then plugging in Equation (2.47), afterwards tayloring the exponential and rewriting.

$$\begin{aligned}
|\chi^\perp\rangle &= \mathcal{N}(|\chi'\rangle - \langle\chi|\chi'\rangle|\chi\rangle), \\
&= \mathcal{N}(e^{-i\hat{B}}|\chi\rangle - \langle\chi|e^{-i\hat{B}}|\chi\rangle|\chi\rangle), \\
&= \mathcal{N}(\mathbb{1} - i\hat{B}|\chi\rangle - \langle\chi|\mathbb{1} - i\hat{B}|\chi\rangle|\chi\rangle), \\
&= \mathcal{N}(-i\hat{B}|\chi\rangle + i\langle\chi|\hat{B}|\chi\rangle|\chi\rangle), \\
&= \mathcal{N}(-i\hat{B}|\chi\rangle + i\langle\hat{B}\rangle_\chi|\chi\rangle).
\end{aligned} \tag{2.49}$$

Consequently, the normalized $|\chi^\perp\rangle$ is given by

$$|\chi^\perp\rangle = \frac{i\langle\hat{B}\rangle_\chi - i\hat{B}}{\Delta B}|\chi\rangle. \tag{2.50}$$

From there on, the beforementioned $\langle\chi|\hat{x}|\chi^\perp\rangle$ can be calculated via

$$\begin{aligned}
\langle\chi|\hat{x}|\chi^\perp\rangle &= \langle\chi|\hat{x}\frac{i\langle\hat{B}\rangle_\chi - i\hat{B}}{\Delta B}|\chi\rangle \\
&= \langle\chi|\hat{x}\frac{i\langle\hat{B}\rangle_\chi}{\Delta B}|\chi\rangle - \underbrace{\langle\chi|\frac{i\hat{x}\hat{B}}{\Delta B}|\chi\rangle}_{=0} \\
&= \langle\chi|\frac{i\epsilon_x\hat{x}\hat{k} - \epsilon_k\hat{x}^2}{\Delta B}|\chi\rangle \\
&= \langle\chi|\frac{i\epsilon_x\hat{x}\hat{k}}{\Delta B}|\chi\rangle - \langle\chi|\frac{-i\epsilon_k\hat{x}^2}{\Delta B}|\chi\rangle \\
&= \langle\chi|\frac{i\epsilon_x(\frac{1}{2}(\hat{x}\hat{k} + \hat{x}\hat{k} - \hat{k}\hat{x} - \hat{k}\hat{x} + \hat{k}\hat{x} + \hat{k}\hat{x}))}{\Delta B}|\chi\rangle + \frac{i\epsilon_k}{\Delta B}\langle\hat{x}^2\rangle_\chi \\
&= \langle\chi|\frac{i\epsilon_x\frac{1}{2}([\hat{x}, \hat{k}] + \{\hat{x}, \hat{k}\})}{\Delta B}|\chi\rangle + \frac{i\epsilon_k}{\Delta B}\langle\hat{x}^2\rangle_\chi \\
&= \frac{\epsilon_x}{2\Delta B} + \frac{i\epsilon_k}{\Delta B}\langle\hat{x}^2\rangle_\chi,
\end{aligned} \tag{2.51}$$

where again the same assumption $\{\hat{x}, \hat{k}\} = 0$ as earlier was made, which is only true near the focus of the beam (cf. section 3.6)

Comparing Equations (2.50) and (2.41) leads to the relationship between ΔB and ϵ

$$\begin{aligned}
\eta\epsilon &= \Delta B, \\
\Delta B &\approx \epsilon,
\end{aligned} \tag{2.52}$$

where $\eta = \frac{1}{\sqrt{1+\epsilon^2}}$ was used. And finally, δx is given by

$$\delta x = \epsilon_x - 2\epsilon_x \text{Re}(\mathbf{P}_B)_w + 4\epsilon_k \Delta x^2 \text{Im}(\mathbf{P}_B)_w. \tag{2.53}$$

2.3 Mach-Zehnder Interferometer

Analogously to δx , a description for δk can be found by first using Equation (2.50) to determine

$$\langle \chi | \hat{k} | \chi^\perp \rangle = \frac{\epsilon_k}{2\Delta B} - \frac{i\epsilon_x}{\Delta B} \langle \hat{k}^2 \rangle. \quad (2.54)$$

Therefore

$$\delta k = \epsilon_k - 2\epsilon_k \operatorname{Re}(\mathbf{P}_B)_w - 4\epsilon_k \Delta k^2 \operatorname{Im}(\mathbf{P}_B)_w. \quad (2.55)$$

More than δx and δk , the total shift δr of the pointer has to be expressed. The following substitutions will be used:

$$\epsilon_x = d, \quad (2.56)$$

$$\epsilon_k = \frac{2\pi}{\lambda} \theta, \quad (2.57)$$

$$z_R = \frac{\pi w_0^2}{\lambda}, \quad (2.58)$$

$$\Delta x^2 = \frac{w_0^2}{4}, \quad (2.59)$$

$$\Delta k^2 = \frac{1}{w_0^2}, \quad (2.60)$$

$$|\mathbf{k}| = \frac{2\pi}{\lambda}, \quad (2.61)$$

with d , the spatial displacement and the angle θ characterizing the angular offset of the beam.

Altogether, with the properties of a Gaussian beam, that the anticommutator $\{\hat{x}, \hat{k}\}$ vanishes near the waist (cf. chapter 3.6), the following formula for δr can be derived:

$$\delta r = \delta x + z\delta\theta = \delta x + \frac{z}{|\mathbf{k}|} \delta k, \quad (2.62)$$

$$\begin{aligned} \delta r = & d - 2d \operatorname{Re}(\mathbf{P}_B)_w + 2z_R \theta \operatorname{Im}(\mathbf{P}_B)_w, \\ & + z\theta - 2z\theta \operatorname{Re}(\mathbf{P}_B)_w - 2\frac{z}{z_R} d \operatorname{Im}(\mathbf{P}_B)_w. \end{aligned} \quad (2.63)$$

This model shall eventually be fitted in the experimental part in order to attain the parameters d and θ with

$$(\mathbf{P}_B)_w = \frac{1 + \cot \alpha e^{-\frac{2\epsilon_x^2}{w_0^2}} e^{-\frac{w_0^2 \epsilon_k^2}{2}} e^{i\phi}}{1 + \cot \alpha^1 + 2 \cot \alpha e^{-\frac{2\epsilon_x^2}{w_0^2}} e^{-\frac{w_0^2 \epsilon_k^2}{2}} \cos \phi} \quad [6]. \quad (2.64)$$

2 Weak Measurements and a Possible Realization

3 Experimental setup

In this chapter, the experimental procedures will be explained. In the beginning, an overview of several optical components will be given. Subsequently, the interferometric setup and its practical implementation will be discussed thoroughly. Afterwards, the actual final setup will be explained. Then, the laser beam will be characterized, and a vital element of the interferometer, the polarizing beam splitter, will be shortly analyzed. Lastly, the measurement will be detailed.

The overall experimental procedure will be the following: First of all, a MZI without a Dove prism will be set up. It will not be perfectly aligned; therefore, the misalignment parameters will be measured through a phase scan, and the MZI iteratively aligned via the method described in section 2.3.1. Once the MZI is perfectly aligned to the extent of our measurement capabilities¹, the Dove prism will be placed in one arm of the interferometer. From then onwards, beam deflections outside the interferometer will be measurable; data will be fitted to Equation (2.63) via a measurement method proposed in [6].

¹The used camera has a pixel size of $4.65\mu\text{m}$. No smaller distances can be resolved for the spatial shift $\delta\mathbf{r}$ and the angular shift $\delta\boldsymbol{\theta} \times \mathbf{z}$.

3.1 Optical Components

In this section, optical components will be reviewed to understand the elements of the interferometer and later the pre-and postselection processes.

3.1.1 Polarizer

Before understanding a Polarizer, one should first remember what polarization is: the spatial expansion of transverse waves.

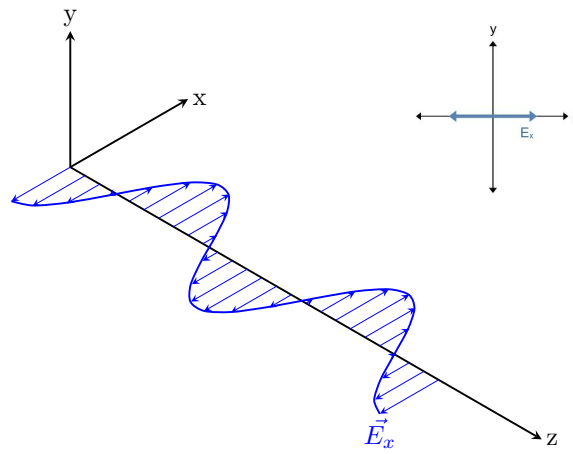


Figure 3.1: Linearly Polarized Electromagnetic Wave.

In electromagnetic waves, the electric field oscillates perpendicular to the direction of travel, as visualised in Figure 3.1.² An electromagnetic wave can be linearly polarized, as in Figure 3.1. This state can more precisely be described as horizontally polarized light, as it oscillates in the x-z-plane. Quantum mechanically, this can be written as the polarization state ket $|H\rangle$.

But an electromagnetic wave can also be elliptically polarized, which is the more general case. If two linearly polarized waves overlap, and they oscillate with the same frequency but in different directions (e.g., along x and y) and they are not in phase, the superpositioned wave is elliptically polarized (circular polarization being the special case when the amplitudes of both waves are the same and the phase shift is $\frac{\pi}{2}$).

In order to illustrate this, imagine a second wave polarized in y-direction, as in Figure 3.2. This polarization state can be expressed as $|V\rangle$ (vertically polarized).

²The magnetic field also oscillates perpendicular to the direction of travel (orthogonal to the electric field). Nevertheless, only the direction of the electric field is significant for polarization – a matter of convention.

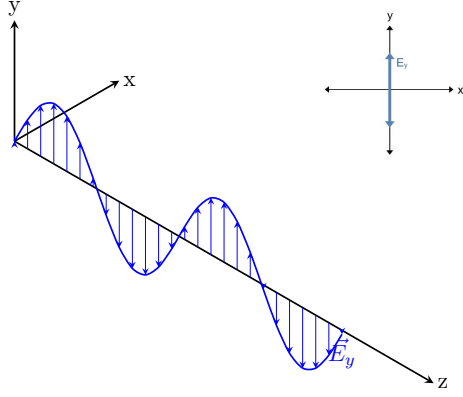


Figure 3.2: Y-Polarized Light.

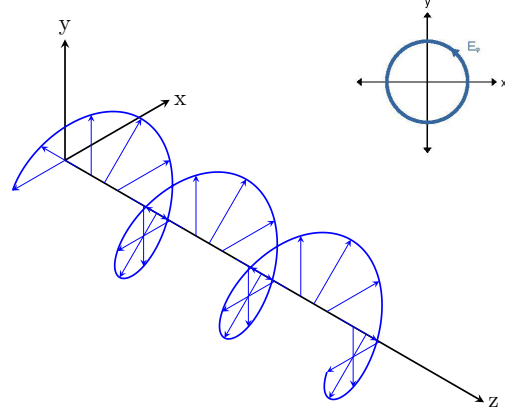


Figure 3.3: Circular Polarized Light.

Note that the electric fields in Figure 3.1 and Figure 3.2 do not oscillate in phase. Suppose we overlap these two linearly polarized waves. In that case, we can create elliptically polarized light, or even a circular polarized electromagnetic wave, if the amplitudes are equal and the phase shift is exactly $\frac{\pi}{2}$ [3]. Circularly polarized light is shown in Figure 3.3. The polarization state in this figure can be characterized by $|L\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle)$. The L stands for left hand polarized light, referring to the anticlockwise rotation of the E-vector. The imaginary unit in the equation represents the phase shift between $|H\rangle$ and $|V\rangle$ of $\phi = \frac{\pi}{2}$, since $i = e^{i\frac{\pi}{2}}$. Likewise, the state vector $|R\rangle$ represents right hand polarized light (clockwise rotation) and $|R\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle)$.

For the sake of completeness two more linear polarization states will be introduced; the states $|+\rangle$ and $|-\rangle$. They characterize an electric field oscillating 'diagonally' or horizontally polarized light tilted by 45° or 135° . The state kets can be rewritten in terms of horizontally and vertically polarized light as $|+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle)$ [30].

What do polarizers do? Intuitively, polarizers filter out one special direction of polarization, illustrated in Figure 3.4. There are different types of polarizers using various effects to achieve this. Some absorptive polarizers are made of dichroic crystals that absorb certain polarizations or are made of a nanoparticle linear film that only transmits one polarization. There are also beam-splitting polarizers that accomplish the same effect by, for instance, reflection at brewster angle or birefringent crystals (that split a beam into two orthogonally polarized beams)

3 Experimental setup

[3, 22]. The polarizers operated in the experiment are absorptive polarizers with a polymer film.

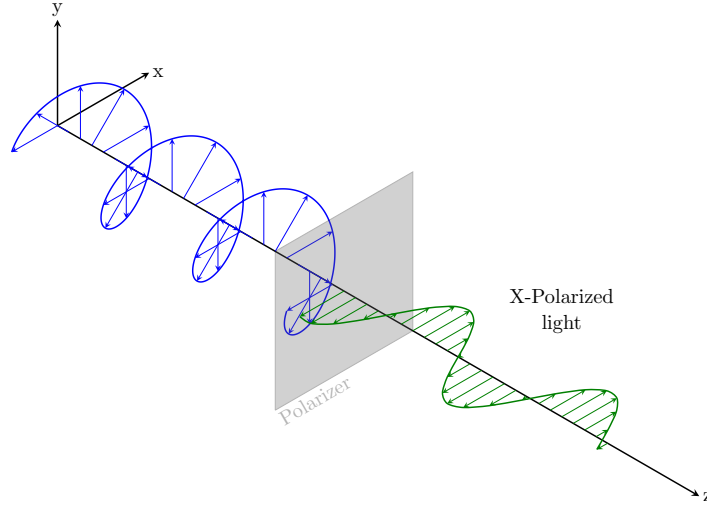


Figure 3.4: Polarizer Filtering out Horizontally Polarized Light.

In the interferometric setup, polarizing beam splitters (PBS) will be used. They work the same as beam-splitting polarizers explained earlier. As will be described in section 3.3.4, most PBSs split into one fully polarized beam and another with a slightly perturbed polarization. Ideally, however, both beams should be fully polarized [22].

In order to quantify how well both beams are polarized, the *contrast* of a PBS gives us a characteristic value. This will later be used to distinguish between the polarizing beam splitters examined for use in the interferometric setup.

The contrast is determined in the following way: Suppose a PBS splits into a horizontally and a vertically polarized beam.³ Then, put horizontally/vertically polarized light through the PBS and measure the intensities at both output ports. Ideally, no intensity should be observed at one of the ports, while the full intensity should be measured at the other port. Nevertheless, experimentally one will observe a tiny intensity at the dark port.

The contrast c is then given by $\frac{I_{\leq}}{I_{>}}$, the ratio between the intensities at both ports. For the purpose of this experiment, the contrast should be less than $\frac{1}{1000}$.

³It works analogously with other polarization states.

3.1.2 Waveplate

Waveplates are made of birefringent materials, i.e., materials with different refraction indices for different linear polarized electric fields along two perpendicular crystal axes. Thus, a phase shift is introduced, which alters the polarization of the incoming beam. The phase shift depends on wavelength, length of birefringent material, and refractive index. Controlling these parameters allows attaining an arbitrary polarization.

Common waveplates are $\frac{\lambda}{2}$ - and $\frac{\lambda}{4}$ waveplates, which rotate the polarization or transform linear polarized light into elliptically polarized one and vice versa [23].

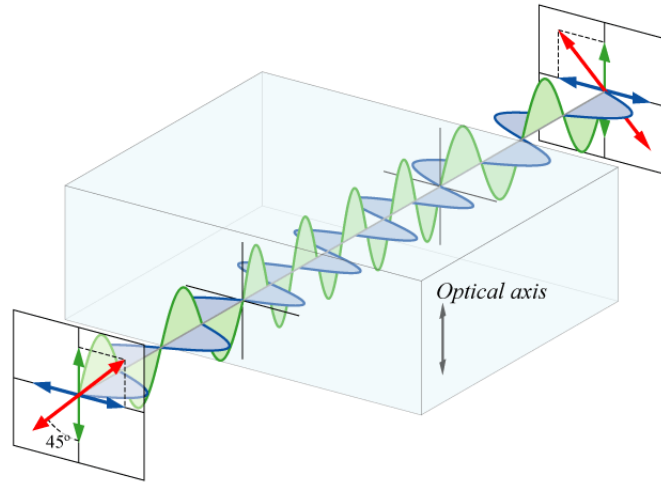


Figure 3.5: Half Wave Plate. Taken from [24].

In Figure 3.5 the incoming linearly polarized wave is split by the half wave plate into two linearly polarized waves parallel and perpendicular to the optical axes of the birefringent material. The perpendicular wave travels through the crystal more slowly. Thus, the phase between both waves is shifted by π , resulting in the outgoing beam's polarization being mirrored on the optical axes.

A quarter wave plate, in contrast, introduces a phase shift of $\frac{\pi}{2}$. This turns polarized light in state $|+\rangle$ into right hand circular polarized light $|R\rangle$ (or analogously $|-\rangle$ into $|L\rangle$).

3.2 Gaussian Beam

The laser used in this experiment radiates an imperfect Gaussian beam. A Gaussian beam is characterized by the Gaussian amplitude envelope along to direction of travel (here the z -axis). It is rotationally symmetric along z , so the mathematical description depends only on the radius r [27]. Additionally, the beam profile is characterized by a Gaussian function. The following figure represents a cross section of a Gaussian beam.

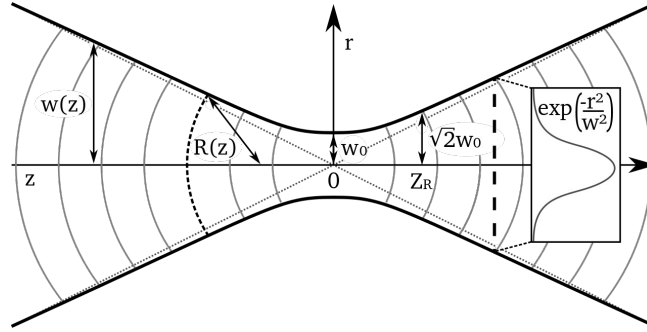


Figure 3.6: Gaussian Beam. Modified from [25].

The field amplitude is given by

$$E_G(z, r) = E_0 \frac{w_0}{w(z)} e^{-ik \frac{r^2}{R(z)}} e^{-i(kz - \eta(z))}, \quad (3.1)$$

with the wave vector $k = \frac{2\pi}{\lambda}$ and the waist w_0 , which is half the beam width at the focus of the beam at $z = 0$. Moreover, $R(z)$ characterizes the radius of the wavefronts (the phase is constant on a wavefront). The Guoy phase is given by $\eta(z)$ and necessary to describe the phase shift by π after the focus. Furthermore, $w(z)$ denotes the waist of the beam. The waist is defined via the radius r where the field amplitude dropped by $\frac{1}{e}$ [26].

The waist of the Gaussian beam can be approximated by:

$$w^2(z) = w_0^2 + M^4 \left(\frac{\lambda}{\pi w_0} \right)^2 (z - z_0)^2. \quad (3.2)$$

Here the focus of the beam lies at $z = z_0$ (in the figure above $z_0 = 0$). In order to quantify how well the beam in the experiment resembles a Gaussian beam, the waist parameter M^2 will be used. M^2 can only take values bigger or equal to 1. If $M^2 = 1$, then the beam is a perfect Gaussian beam which is not achievable experimentally. However, the value should be close to 1 [27].

This value will later be determined to characterize the laser in this experiment.

Furthermore, the Gaussian beam can be characterized by a certain condition around the focus position: $\{\hat{x}, \hat{k}\} = 0$. This simply means that at the waist, the position x of a photon cannot be definitely characterized by its momentum k and vice versa. At the focus, all momentum vectors (pointing to the right) are parallel to each other and the wavefront is plane. The momenta at each point are orthogonal to the wavefront, the momentum vectors before the focus aim at the waist, while afterwards they diverge.

This assumption was vital to the calculations, where $\{\hat{B}, \hat{O}\} = 0$ was often used in Equations (2.11), (2.33), (2.46), (2.51), where this relation was already indicated.

3.3 Interferometer

3.3.1 Analysis of Interferometric Setups

In this section, the interferometer setup will be explained and how and why it differs from the originally planned interferometer, characterized in the theoretical part of this thesis.

Foremost, assembling a Mach-Zehnder interferometer is a relatively intricate task because it is not very stable. Small imprecisions in the build-up or environmental factors can introduce phase shifts, which spoil interference patterns.

For this reason, a Sagnac interferometer was built. The great advantage is that both beams pass through the same optical components and the exact same path, making it simpler, compact, and stable. In Figure 3.7 one such interferometer is depicted. The beam splitter divides the incoming beam; both beams are reflected by three mirrors and go back to the beam splitter, where they are again reflected or transmitted and detected at the output port.

Since both beams take the exact same path, there is no relative phase shift ϕ between the two beams in a perfectly aligned Sagnac interferometer. In practice, however, at least one mirror is slightly shifted, and therefore both beams are spatially separated. Furthermore, at least one mirror is a little tilted at a small angle, so the beams interfere because of an angle between them. Determining the phase shift ϕ is not only essential for understanding the interference pattern but also crucial to computing the weak value (cf. Equations (2.32), (2.38) and (2.64), (2.63)). How can we determine the phase shift in a MZI or a Sagnac interferometer?

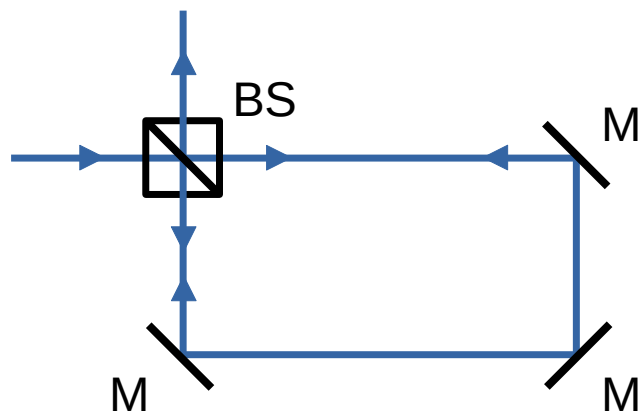


Figure 3.7: Sagnac Interferometer.

In a Mach-Zehnder interferometer, the phase shift can be controlled by changing the path length of one arm or inserting a medium in one arm. The changing of path length can be conveniently performed in a folded MZI, as shown in Figure 3.8, where one can lengthen or shorten one arm of the beam by moving the prisms. However, in the Sagnac interferometer, no such expansion of one arm is possible, as moving the mirrors (cf. Figure 3.7) would result in stretching the other beam's path as well, which would not generate a relative phase.

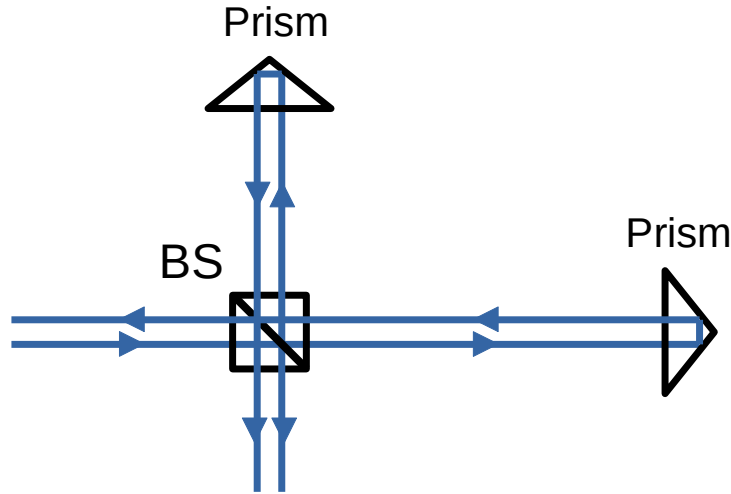


Figure 3.8: MZI with Extensionable Arms.

In order to solve this issue, one first has to spatially separate both beams in the Sagnac interferometer. This is performed by slightly tilting one mirror – this is usually the case when the Sagnac interferometer is not perfectly aligned. Then one can insert a medium, for example, a glass platelet, into one arm. The laser beam passing the small glass plate is slowed down because glass is denser than air (cf. refraction of light), and therefore the beam's phase is altered. By tilting the glass plate, the length of the beam's path inside the glass medium can be modified, and thus the phase shift ϕ varied.

Unfortunately, the beam would also be slightly shifted (if it does not hit the glass plate at exactly 90°), as demonstrated in Figure 3.9, which is practically never the case because for the different phase shifts, the glass plate has to be tilted.

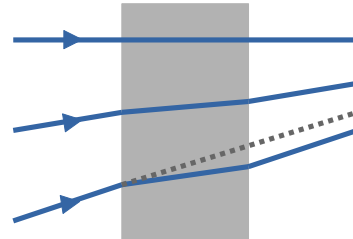


Figure 3.9: Beam Path in Glass Platelet.

3 Experimental setup

Since it would experimentally be more convenient not to generate another spatial shift of the beam that will have to be taken into account when measuring the final beam deflections, another method has to be considered.

It is namely, tuning the phase shift by polarizing the laser beam. This can be done in the following way.

Imagine the system to be in a coupled state, describing photons in arm A as horizontally polarized and photons in arm B is vertically polarized. Instead of having the state $|\psi\rangle$ that described the Misaligned MZI $|\psi\rangle = \cos(\alpha)|A\rangle|\chi'\rangle + \sin(\alpha)e^{i\phi}|B\rangle|\chi\rangle$ (cf. Equation (2.25)), we have $|\psi_{Pol}\rangle$ where arm A is not only coupled to the shifted pointer state $|\chi'\rangle$ but also to the horizontal polarization $|H\rangle$. In the same way, arm B is coupled to $|\chi\rangle$ and $|V\rangle$. The polarized state is expressed by

$$|\psi_{Pol}\rangle = \cos(\alpha)|A\rangle|\chi'\rangle|H\rangle + \sin(\alpha)e^{i\phi}|B\rangle|\chi\rangle|V\rangle. \quad (3.3)$$

The postselection process consists of two projections. Firstly, the state $|\psi_{Pol}\rangle$ is projected onto output port C via $\langle C | \psi_{Pol} \rangle$ – as analogously done in Equation (2.27) – and further onto the polarization $|H\rangle + e^{i\phi}|V\rangle$. This leads to the same final pointer state as in Equation (2.27).

$$\begin{aligned} |\psi'_F\rangle &= \cos(\alpha)|\chi'\rangle + \sin(\alpha)e^{i\phi}|\chi\rangle, \\ &= \mathcal{N} \left(|\chi\rangle + \frac{\eta\epsilon}{\eta + \tan(\alpha)e^{i\phi}}|\chi^\perp\rangle \right), \end{aligned} \quad (3.4)$$

where in the last transformation Equation (2.41) was used.

The projection onto $|H\rangle + e^{i\phi}|V\rangle$ can be experimentally implemented by firstly positioning a quarter wave plate (QWP) at a 45° angle, which converts linearly polarized light into circularly polarized light and secondly placing a polarizer behind the QWP. Together these two components project onto all states in the $\sigma_{x,y}$ plane of the Bloch-sphere spanned by the basis kets $|R\rangle$, $|L\rangle$, $|+\rangle$ and $|-\rangle$.

If $\phi = 0$ or $\phi = 2\pi$ then the QWP and the polarizer act as a projection onto $|+\rangle$ ($|H\rangle + e^{i\cdot 0}|V\rangle = |+\rangle$). For $\phi = \frac{\pi}{2}$, however, one projects onto $|L\rangle$. $\phi = \pi$ generates light in state $|-\rangle$ and $\phi = \frac{3\pi}{2}$ right hand polarized light $|R\rangle$. The different polarization states written in the $\{|H\rangle, |V\rangle\}$ basis can be reread in section 3.4.

How can the horizontal and vertical polarization of both beams be implemented? One can either position polarizers in both beam paths (after the beam splitter) as seen in Figure 3.10 or directly use a polarizing beam splitter, demonstrated in Figure 3.11.

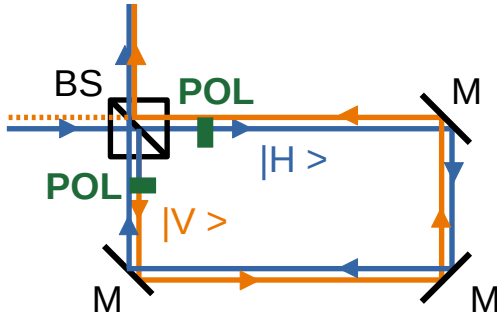


Figure 3.10: Sagnac Interferometer with Polarizer in Each Arm.

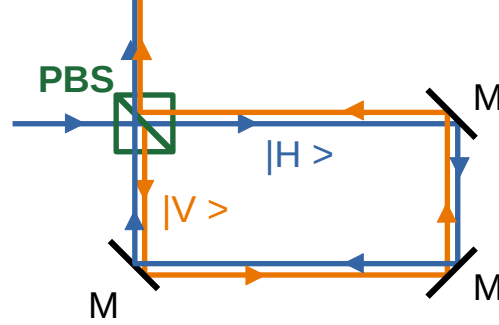


Figure 3.11: Sagnac Interferometer with Polarizing Beam Splitter.

Both structures have their pro and contra. The first method (Figure 3.10) lacks that a lot of beam intensity is lost in the interferometer. Suppose the laser beam is in the polarization state $|+\rangle = |H\rangle + |V\rangle$, normalization omitted. The incoming beam is split 50:50 by the beam splitter⁴ and then both beams pass the polarizers. Suppose the polarizers extract $|H\rangle$ and $|V\rangle$ polarizations. Then, half of the intensity is cut off because either $|H\rangle$ or $|V\rangle$ are transmitted. Afterwards, the beams go through the same beam splitter again, where around 50% of the beam is lost because we only measure the output port at the top. Altogether $\frac{1}{4}$ of the initial intensity is ideally left.

Intensity loss per se is not an issue for the experiment, as the laser used in the laboratory is able to deliver much higher intensities than the ones worked with for this experiment. Yet, if this experiment turns into a long-term project, then energy efficiency certainly is an important criterion. Furthermore, placing the polarizers in a position such that the mount of the polarizer would not block the other beam's path⁵ was not practicable without a major modification of the polarizer's mounts, namely cutting off roughly one-fourth of the mount.

On the other hand, polarizing beam splitters have their deficiencies. Although they should purely reflect one polarization and wholly transmit another, different results were observed experimentally. This will be expressed in more detail in section 3.3.4.

⁴Realistically viewed, neither the ratio 50:50 is perfectly achieved nor the transmissivity of the beam splitter is fully 100%.

⁵Remember that for a not ideal Sagnac interferometer, the beams pass through slightly different paths because at least one mirror is slightly displaced.

3 Experimental setup

After various considerations expressed in the previous pages, the last interferometric setup (Sagnac interferometer with a PBS, cf. Figure 3.11) seemed to be the most advantageous and was chosen to be built. More will follow in the next section.

3.3.2 Final Interferometer Structure

The final setup of the interferometer is visualized in the following figure. The distances of the components are also marked in the figure; however, they are error afflicted ($\pm 0.5\text{cm}$). Two mirrors in the front enable not only fixing the height of the incoming beam but also the horizontal shift of the beam and the deflection angle θ . This will later be used to deflect the beam outside the interferometer. Nevertheless, for the alignment of the setup, they were positioned to transmit the beam with almost no shift (parallel to the optical table and along a row of holes on the table).

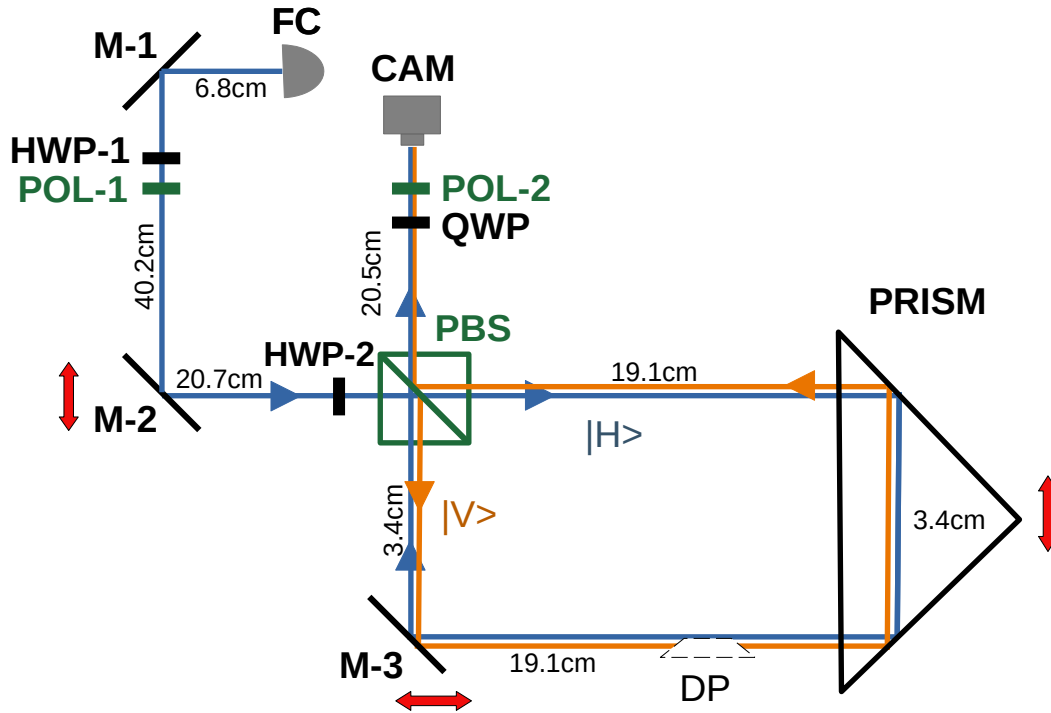


Figure 3.12: Final Interferometer Setup.

The transmitted and reflected beam of the interferometer (marked as $|H\rangle$ and

$|V\rangle$) do not propagate along the exact same path⁶. In fact, they are purposely separated by moving the third mirror (M-3) to position the dove prism in one of the arms in order to implement the beam deflection measurement described in chapter 2.7.

The prism on the right differs from the setups characterized in section 3.3.1. Instead of operating two mirrors, it is more advantageous to use one prism because it is more compact, and the 90° angle between both reflecting layers is already set precisely.

The small trapezium on the bottom right represents the future position of the Dove prism. But for the alignment method it was not set up yet.

The red arrows in Figure 3.12 indicate the movable components in this setup. These are the degrees of freedom of the beam. Each of these movable components is built on shiftable tables. The components can be translated one-dimensionally and finely tuned by micrometer screws.

Moving the second mirror (M-2) up or down results in moving the beam up or downwards (from a bird's-eye perspective). In the interferometer, this would move both beams up or down. In the coordinate system presented earlier (z-axis along beam, x-axis: horizontal plane (parallel to the table), y-axis: vertical plane (orthogonal to the table surface)), this movement would be interpreted as the beam translating along the x-axis.

The third mirror can also be displaced. By moving it to the left, the reflected beam $|V\rangle$ moves outwards, away from the H-polarized beam, and vice versa when shifting it to the right.

Furthermore, by moving the prism up or down (from a bird's view), both beams can drift apart or come closer until they overlap and afterwards flip positions (horizontally polarized beam being outside and vertically polarized beam being inside).

Apart from that, the setup also includes half and quarter wave plates and polarizers. The first polarizer⁷ ensures that the beam's polarization coming out of the fiber coupler stays constant. Sometimes the movement of the single-mode fiber can change the beam's polarization state. Yet, this has to be avoided because it would change the ratio of how much of the beam is transmitted or reflected by the polarizing beam splitter and thus alter the α value in Equations (2.25) and (2.42), which is crucial to the derived equations of the pointer shift (see Equations (2.38), (2.63)). A steady polarization could similarly be achieved by putting bat ears on the single-mode fiber. They are also often employed to fix polarizations. The first half wave plate (HWP-1) in front of the first polarizer (POL-1) rotates the polarization state. Nonetheless, since the first polarizer right after HWP-1 only transmits V-polarized light, rotating the half wave plate is equivalent to ad-

⁶Still, both beams have the same path length!

⁷In the experiment, a V-polarizer was utilized. But an arbitrary polarization can be chosen.

3 Experimental setup

justing the beam's amplitude.

The second half wave plate helps to change the beam from state $|V\rangle$ (vertically polarized beam) into a desired mix of horizontally and vertically polarized light ($\cos(\alpha)|H\rangle + \sin(\alpha)|V\rangle$). The polarizing beam splitter will then transmit and reflect certain intensities (e.g., 50:50 ratio at $\alpha = 45^\circ$), wherefore the intensities in both arms can be controlled mutually (the sum of the intensities is constant due to energy conservation).

After both differently polarized beams pass the interferometer and exit at the top, there is no interference before they transit the quarter wave plate and the following polarizer because the horizontally and vertically polarized beams cannot interact. At the quarter wave plate, both beams are polarized circularly. Afterwards, the polarizer projects onto a chosen linear polarization (cf. chapter 3.3.1 again for a more detailed mathematical approach). We then measure this interfering state with the camera at the top end.

3.3.3 Laser Beam Setup

The first task of the experiment was setting up the laser beam. An infrared beam with an 810nm wavelength was used, whose wavelength is just outside the visible electromagnetic spectrum. Yet, the beam could be made visible using a laser viewing card (IR-card) that absorbs the beam and radiates visible light because of a fluorescent film on the card.

In the experimental setup, a laser diode's outgoing beam was directed into a single-mode fiber, leading to a fiber coupler (FC). The fiber coupler was then used to collimate the laser beam with a built-in movable lense. However, the beam cannot be collimated perfectly since the laser's output is a Gaussian beam with a focus, as described in section 3.2. Nonetheless, the best collimation degree has to be achieved for the assumptions to be true that $\{\hat{x}, \hat{k}\} = 0$ (cf. for example Equations (2.51), (2.63)), as made in the calculation. Thus, the diverging light exiting the single-mode fiber was roughly collimated. The differing beam width could be observed by moving the IR card along the laser beam. Since the detector will afterwards be positioned at 1m, the focus had to be set there. This could not be done with the bare eye and the laser viewing card. Therefore a camera (IDS CM OS camera) was placed in various positions, and from the beam profiles the widths were determined. Through iteratively taking pictures, evaluating the focus position (at the smallest waist), and moving the lense, the focus was set at 1m.

In addition, two micrometer screws on the mount of the fiber couple were turned

to rotate the FC around a vertical axis and a horizontal axis (cf. Figure 3.13). In this way, the propagation direction of the beam was set. With two irides (cf. Figure 3.14), one close to the FC, and one very far away (around 1.5m), the beam was fixed to radiate along a straight line along the horizontal plane (parallel to the optical table) and had no spatial shift to the left or right (x-direction). In other words, the beam propagated along a row of holes on the optical table without an angle in the y-direction (perpendicular to the table surface).



Figure 3.13: Mount of Fiber Coupler and Mirrors.
Taken from [28].



Figure 3.14: Iris. Modified from [29].

Once this was done, a more precise focus measurement could be carried out using the camera again. Various pictures of the beam profiles were taken along the beam path. This was done to determine the position of the waist z_0 (cf. section 3.2) and to find out whether the laser beam used in the experiment was a good approximation of a Gaussian beam. Therefore, the waist had to be fitted using Equation (3.2) to compute M^2 , as presented in section 3.2. In total, 15 pictures were taken of the beam, illustratively shown in Figure 3.15. The first image was taken right after the fiber coupler; another eight were taken near the beam's focus and further six after one Rayleigh length (1.75m).

Then, the photographed beam profiles were analyzed, intensities summed in x and y-direction (cf. Gaussian functions in Figure 3.16), and the waists extracted via the standard deviation.

3 Experimental setup

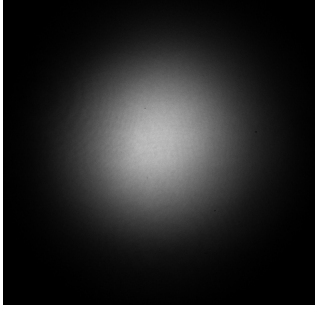


Figure 3.15:
Exemplary Beam
Profile.

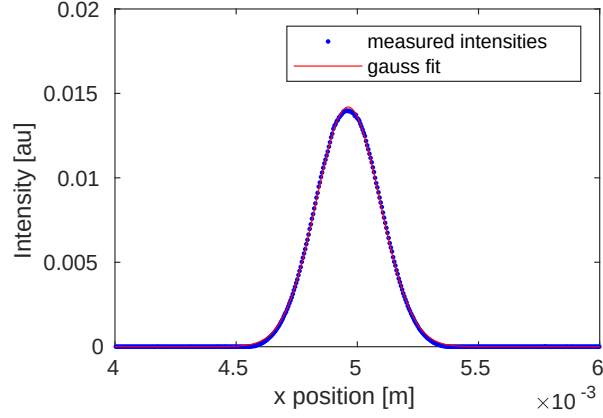


Figure 3.16: Intensity Cross-Section of Figure 3.15.

After the waists were determined at different z positions, they were visualised (Figure 3.17) and the data points fitted with the following equation, already presented in section 3.2.

$$W^2(z) = w_0^2 + M^4 \left(\frac{\lambda}{\pi w_0} \right)^2 (z - z_0)^2. \quad (3.5)$$

The following values were obtained: the waist at the focus $w_{0,x} = 670.2\mu m$, $w_{0,y} = 672.2\mu m$, the focus position $z_{0,x} = 1135.8mm$, $z_{0,y} = 1141.5mm$ and the value for $M_x^2 = 0.93$, $M_y^2 = 0.95$ ⁸.

This M^2 result, however, is not physical since M^2 cannot lie below 1, as put forth in [27]. The lowest value M^2 can attain is 1, representing a single-mode TEM₀₀ Gaussian beam.

However, considering the error interval emerging from the measurement, the M^2 value surely must lie above 1. One has to take into account that background noise, camera digitization, camera non-linearity, and baseline drift can result in measurement inaccuracies. Different cameras in the laboratory were tested; nevertheless, the one worked with seemed to be the most precise. Also, in the future versions of this experiment, carried out by Versmold and Köster, a better detector will be employed, with higher resolutions than the $4.65\mu m$, that might not have the previously mentioned inaccuracies.

Nonetheless, the measurement shows that the beam is well collimated, which is important for the assumptions made in the calculations to hold true.

⁸The deviation is quantified by the root-mean-squared: $rmsq_x = 6.3e - 06$ and $rmsq_y = 6.8e - 06$

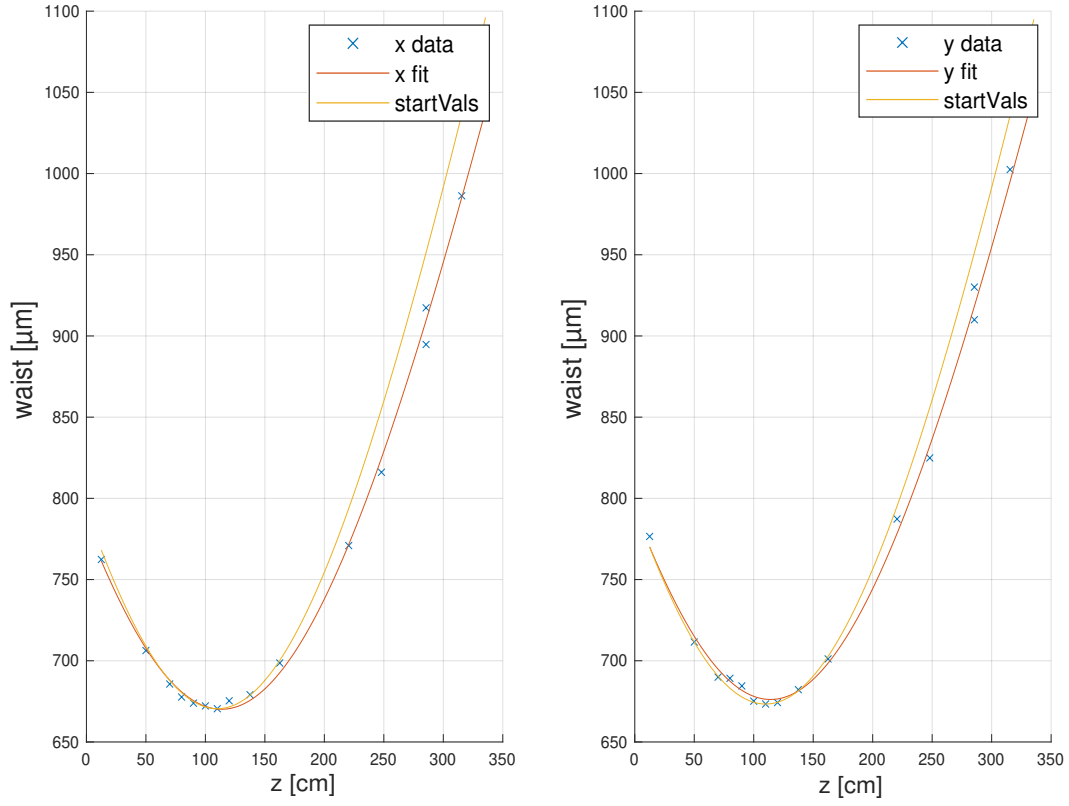


Figure 3.17: Beam Waist Fit.

Moreover, the focus position was determined in the same fit (cf. Equation (3.2)), which is the essential parameter for computing the z-position in the pointer shift $\delta \mathbf{R}$ and δr (cf. Equations (2.38), (2.63)).

3 Experimental setup

3.3.4 Characterizing Polarizing Beam Splitters

The polarizing beam splitters present in the laboratory demonstrated some imperfections or unexpected misbehaviors. But since a PBS is essential to the setup, they were analyzed in more detail. Most of them⁹ had a bad contrast for the H polarization. And some exhibited an angularly dependence. An analysis of these characteristics will be given on the next pages.

The following experimental procedure was conducted to examine the contrast. Firstly, the incoming beam was linearly polarized (horizontally or vertically). Secondly, the PBS was positioned so that the transmitted beam was polarized horizontally and the reflected beam vertically. This was done by rotating the PBS around a horizontal and a vertical axis (cf. Figure 3.13) and minimizing the intensity of the reflected (or transmitted) beam when the incoming beam was horizontally polarized (or vertically polarized).¹⁰ This alignment corresponds to setting the eigenbasis of the PBS to the polarization states $|H\rangle$ and $|V\rangle$. And lastly, the intensities at both PBS outputs were compared for a purely V-polarized incoming beam or a purely H-polarized one. Ideally, one expects only to detect something at one output. If the initial beam was polarized horizontally, one should only measure a transmitted beam because the PBS transmits H-polarization and reflects V-polarization. Analogously, only a reflected beam should be measured for initially V-polarized light. Nevertheless, this was not the case and a small intensity was detected in the other arm.

The ratio between the small and great intensity of the two arms should be below $\frac{1}{1000}$ for our measurement purposes. The following table represents the measured contrasts of a few polarizing beam splitters tested.

	PBS-57	PBS-22	PBS-54	PBS-40
$ H\rangle$	$< \frac{1}{1000}$	$\approx \frac{1}{1000}$	$\approx \frac{2}{1000}$	$\approx \frac{3}{10000}$
$ V\rangle$	$< \frac{5}{1000}$	$\approx \frac{1}{1000}$	$\approx \frac{1}{1000}$	$\approx \frac{2}{10000}$

Table 3.1: Measured Contrasts of Polarizing beam splitters for each Polarization

On a side note, the laboratory's polarizers are inventoried and thus named this way. Since we are looking for PBSs with a contrast better than $\frac{1}{1000}$, the table shows us that only two PBSs have a contrast equal or better in both arms. Naturally, PBS-40 was chosen for the experimental setup.

⁹In total eight polarizing beam splitters were tested.

¹⁰Aligning the minima is a more precise experimental procedure than detecting the maxima.

Nevertheless, later in the experiment, PBS-40 revealed a slight fluctuation of the transmission-reflection ratio over time, although the polarizing beam splitter was not moved. The ratio should be constant throughout the experiment and ideally the PBS should transmit and reflect the same intensities (50:50). A half wave plate in front of the polarizing beam splitter firstly set the ratio to 50.16:49.84. This unequal intensity distribution in the experiment was expected and is represented by the terms $\cos(\alpha)$ and $\sin(\alpha)$ in Equation (2.21). The following table demonstrates the ratios found for a series of measurements approximately one minute apart from each other.

I_{Arm1}	0.4979	0.4949	0.4887	0.4931	0.4989
I_{Arm2}	0.5021	0.5051	0.5113	0.5069	0.5011

I_{Arm1}	0.5040	0.5076	0.4904	0.4972	0.4976
I_{Arm2}	0.4960	0.4924	0.5096	0.5028	0.5024

Table 3.2: Relative Intensity of Both PBS Arms

The most likely reason for these variations is an intensity fluctuation of the laser itself since the measurements on each arm did not occur simultaneously. This could be dealt with by installing a BS and a photodiode in front of the interferometer to measure the incoming beam (cf. Figure 3.18) – as will be done in the future of the experiment. Thus, a small portion of the laser beam would be reflected by the BS and detected by the photodiode.

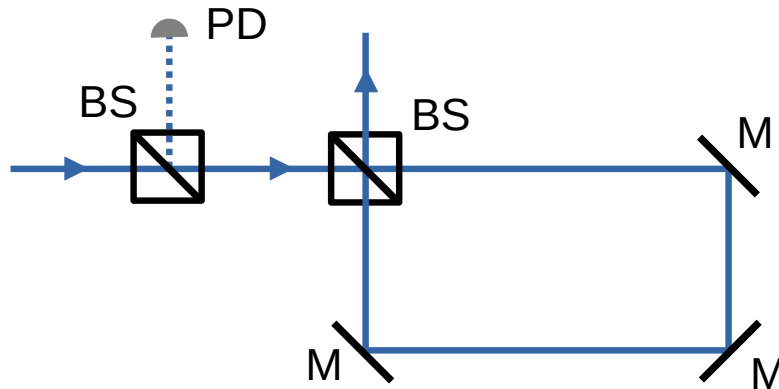


Figure 3.18: Beam Splitter and Photodiode in Front of Sagnac Interferometer to Measure Relative Intensity Fluctuations.

3 Experimental setup

Hence, intensity fluctuations could be observed and filtered out of all measurements afterwards.¹¹

Nonetheless, the ratio values from Table 3.2 are crucial for the determination of α in Equation (2.25) and (2.42). Therefore the average values for the relative intensities were taken, $\bar{I}_{Arm1} = 0.49703$ and $\bar{I}_{Arm2} = 0.50297$ and α computed as

$$\tan(\alpha) = \sqrt{\frac{0.49703}{0.50297}} = 0.99408 \rightarrow \alpha = 44.8298^\circ. \quad (3.6)$$

The error interval of $\tan(\alpha)$ can be computed via Gaussian Error Propagation [32].

$$\Delta \tan(\alpha) = \sqrt{\left(\frac{\partial \tan(\alpha)}{\partial I_{Arm1}} \Delta I_{Arm1}\right)^2 + \left(\frac{\partial \tan(\alpha)}{\partial I_{Arm2}} \Delta I_{Arm2}\right)^2}, \quad (3.7)$$

$$\Delta \tan(\alpha) = 0.027,$$

where $\Delta I_{Arm1} = \Delta I_{Arm2} = 0.0189$. Therefore,

$$\tan(\alpha) = 1.00 \pm 0.03. \quad (3.8)$$

Lastly, many beam splitting effects depend on the angle of incident; the following features of three beam splitters were found. BS-27's reflection-transmission ratio was observed to be dependent on the angle of incidence of the incoming beam and the recorded intensity distribution is visualized in the following Figure 3.19.

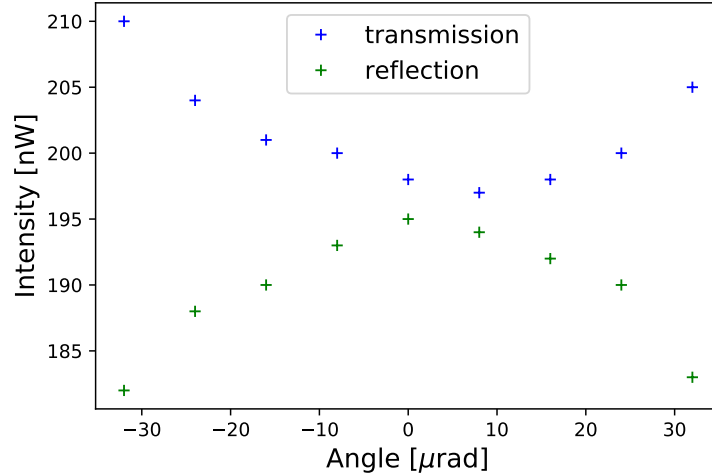


Figure 3.19: Transmitted and Reflected Beam's Intensity Dependent on Angle.

¹¹Another beam splitter for this purpose was already set up, however a photodiode is still missing in the laboratory.

The angle 0 refers to the approximated angle where the ratio is optimal. There the ratio is almost 50:50. When turning the BS, however, the ratio worsens. The angles are achieved through turns of the micrometer screws on the BS mount that rotate the BS around a vertical axis through the beam splitter.

This angular dependence could be explained by the structure of the thin film inside the beam splitter and its reflection and transmission properties; however, no further research was done to determine this issue.

Furthermore, two polarizing plate beam splitters, BS-54 and PBS-57, also depended on the incident angle. However, not the general transmission-reflection ratio, but their contrasts were altered with the changing angle. BS-54's contrast ($< \frac{1}{1000}$) was best at an angle of $\approx 15^\circ$. And the optimal angle for PBS-57 was at $\approx 40^\circ$. Here no further data was collected, then the approximation of the angle at which the contrast was best since later, another beam splitter (BS-40) was used for the final setup due to its very high contrast - BS-40's contrast also differed for various angles; however, the angle had to be relatively big to detect a noticeable difference in the contrast, wherefore this effect could be neglected.

3 Experimental setup

3.3.5 First Observed Interference

Once the laser beam was set up, the optical components put into place, and the right polarizing beam splitter chosen and positioned, the first camera picture could be taken to observe the beam at the output port.

One of the first pictures taken resembled something similar to the following Figure 3.20. Two separate beams can be seen that are set apart by some spatial shift d and some angle θ .

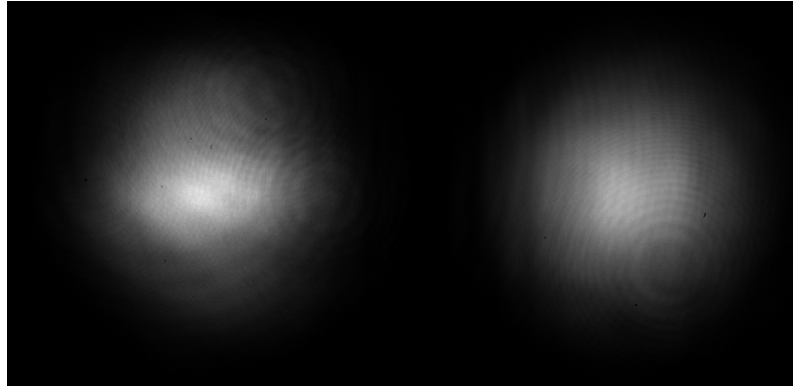


Figure 3.20: Detecting Seperate Beams

There should be no spatial shift, however, due to some tilt in the third mirror, the beams are separated. The tilt in M-3 causes the vertically polarized beam to be deflected early on its path through the interferometer and the horizontally polarized beam to be deflected later. Since the angular displacement is amplified linearly, the further the beam travels, the $|V\rangle$ -beam is displaced more strongly. Therefore both beams do not meet at the same point on the PBS again, and this separation can be detected via the camera.

The third mirror was then tilted. In this way, both beams could be moved together. Once they overlapped no other angle should be seen because, theoretically, both beams should get the same angles from the structure of the setup. However, due to imperfections of optical components (e.g., inaccuracies of reflective or transmissive films of components), an angle was observed between the beams which caused interference fringes (cf. section 2.3.1) when the beams overlapped (see the following Figure 3.21).

If we want to determine this unexpected angle, the alignment method has to be applied. Thus, the measurement procedures could be started (more detail in section 3.4).

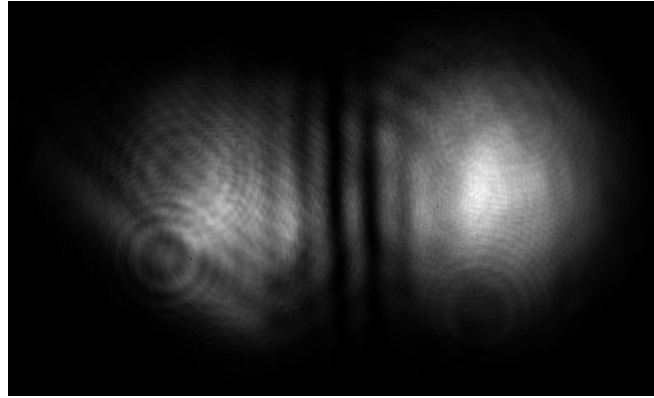


Figure 3.21: Interference Fringes Arising when Moving Beams Closer Together.

One peculiarity was detected: strange interference patterns in one single beam arm. When the contrast of the camera was increased (compared to the contrast in Figure 3.20), the reflected beam of the polarizing beam splitter (vertically polarized) demonstrated horizontal interference lines. The actual photo of the beam is shown in the next figure.

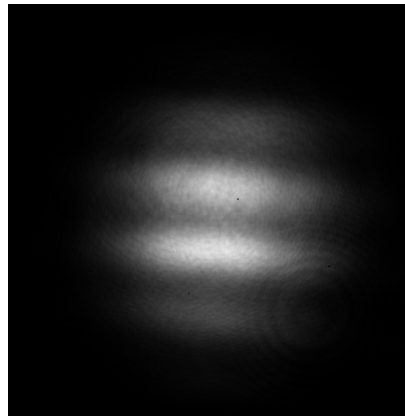


Figure 3.22: Irregularities of Reflected Beam

This affects the interference pattern of both beams and, therefore, should be avoided. In a future setup, another beam splitter will be put into place that does not generate an interference pattern on the reflected beam.

3.4 Measurement Results

In the experiment, the alignment parameters of the interferometer were computed via a phase scan. In the future, the Dove prism will be put into place. This work shall be continued by Versmold, my advisor, and Köster, who Versmold also advises in her master thesis (and who I have mentioned already several times because of her bachelor thesis [6]).

The following measurement was done in order to obtain the misalignment parameters. Several pictures were taken with the camera while rotating the second polarizer - the phase was scanned. Next, the centroids of the beam profiles were extracted and plotted. The centroids' positions produced an ellipse as expected.

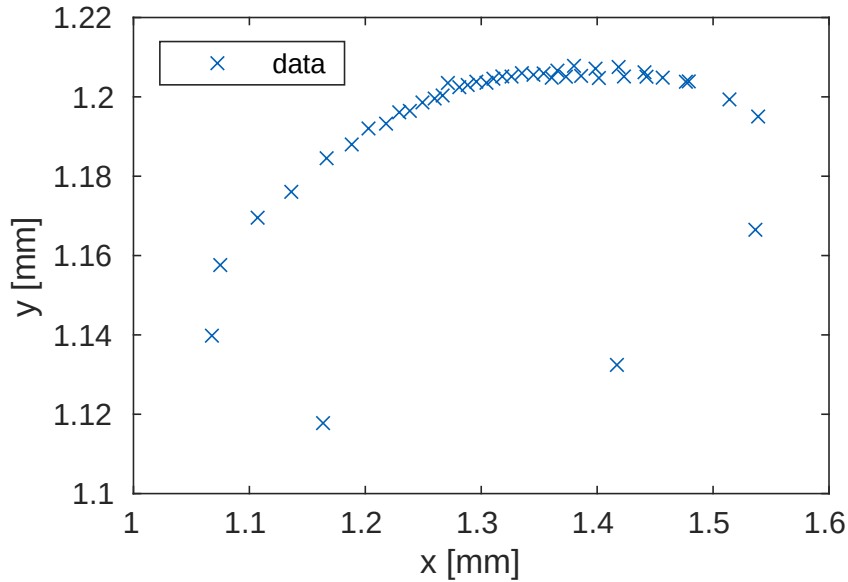


Figure 3.23: Centroids Obtained from Phase Scan.

A few crosses on the top overlap because the polarizer was rotated a little more than 180° , which corresponds to one period $\phi = 2\pi$.

The next figure pictures the intensity distribution dependent on the polarizer's angle ϕ , which was obtained from the same set of data. The curve fitted was

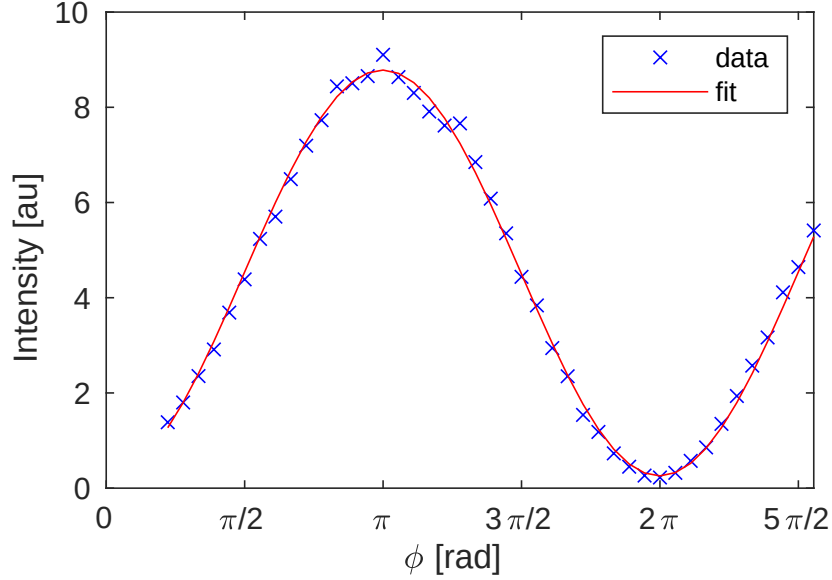


Figure 3.24: Intensity Profile Obtained from Phase Scan.

the expected intensity function $2 \sin(\frac{\phi-\phi_0}{2})^2$ from Equation (2.18). This curve was used to determine the offset ϕ_0 of the phase shift in $e^{i(\phi-\phi_0)}$. ϕ in this formula does not initially correspond to the rotational state ϕ of the polarizer since it is in an arbitrary position in the beginning. Therefore, the offset ϕ_0 has to be computed.

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The centroid's position along the x and y axes can also be visualized in dependence of the phase ϕ . Fitting this data using Equation (2.38) and (2.39) yields the misalignment parameters we are looking for.

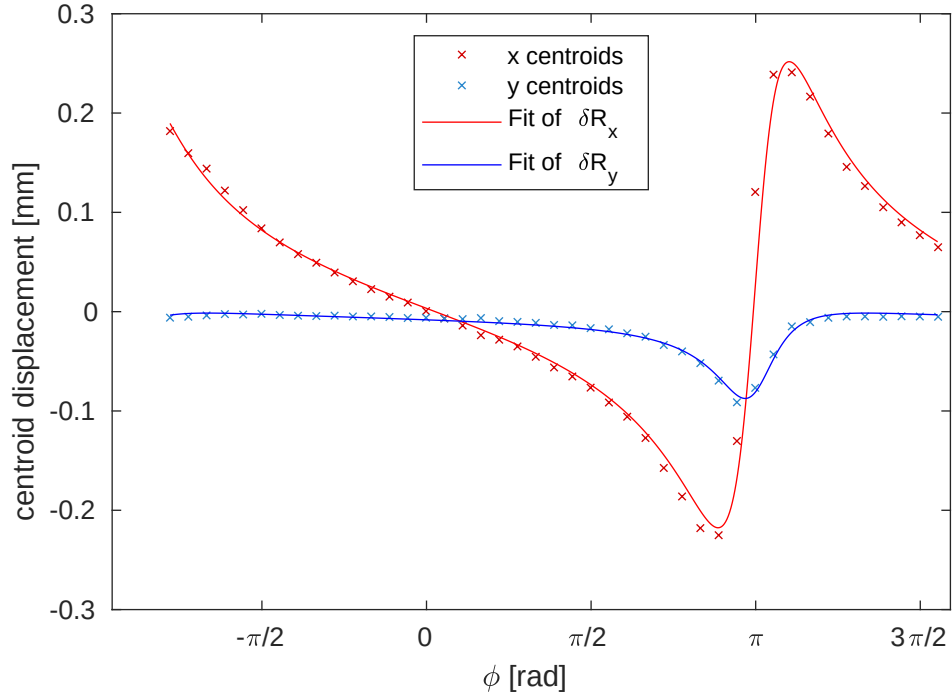


Figure 3.25: X and Y Coordinates of Centroids Dependent on Phase ϕ .

The resulting misalignment parameters are:

$$\begin{aligned}\delta\theta_{x,P,1} &= -46.23 \mu\text{rad}, \\ \delta\theta_{y,P,1} &= 145.8 \mu\text{rad}, \\ \delta_{x,P,1} &= 0.5828 \text{ mm}, \\ \delta_{y,P,1} &= -2.61 \text{ mm}, \\ z_{R,x,1} &= 1707 \text{ mm}, \\ z_{R,y,1} &= 1759 \text{ mm},\end{aligned}$$

where again the error is quantified by the root-mean-squared values:

$\text{rms}_{x,1} = 0.0205$ and $\text{rms}_{y,1} = 0.0028$. The index P stands for phase scan and the index 1 for the first fit.

A first glance at these numbers might let one see that the Rayleigh length obtained from this fit is in great agreement with the computed Rayleigh length using the

beam waist from chapter 3.3.3 and Equation (2.59).

However, one should be suspicious about why the displacement in y-direction $\delta_{y,P,1} = -2.61 \text{ mm}$ is that great. If the laser beam used has a waist of $\bar{w}_0 = 671.2 \mu\text{m}$ (cf. section 3.3.3), the beam's diameter is roughly 1.4mm wide. A displacement of -2.61 mm would separate both beams and therefore make two dots (cf. Figure 3.15) visible in the camera photo, which was certainly not the case.

Nevertheless, let us compare the value fitted for θ and measurements further away from the current camera positions. The shift of both beams due to the angle should increase linearly with the distance to the focus. Pictures at various z-positions were analyzed by Köster to determine the distance between the beams at various positions and thus the angle. The assumption was made that for a small angle θ , the following relation holds for the distance between the beams Δ_{Beams} and the position in z.

$$\theta \approx \tan(\theta) = \frac{\Delta_{Beams}}{z}. \quad (3.9)$$

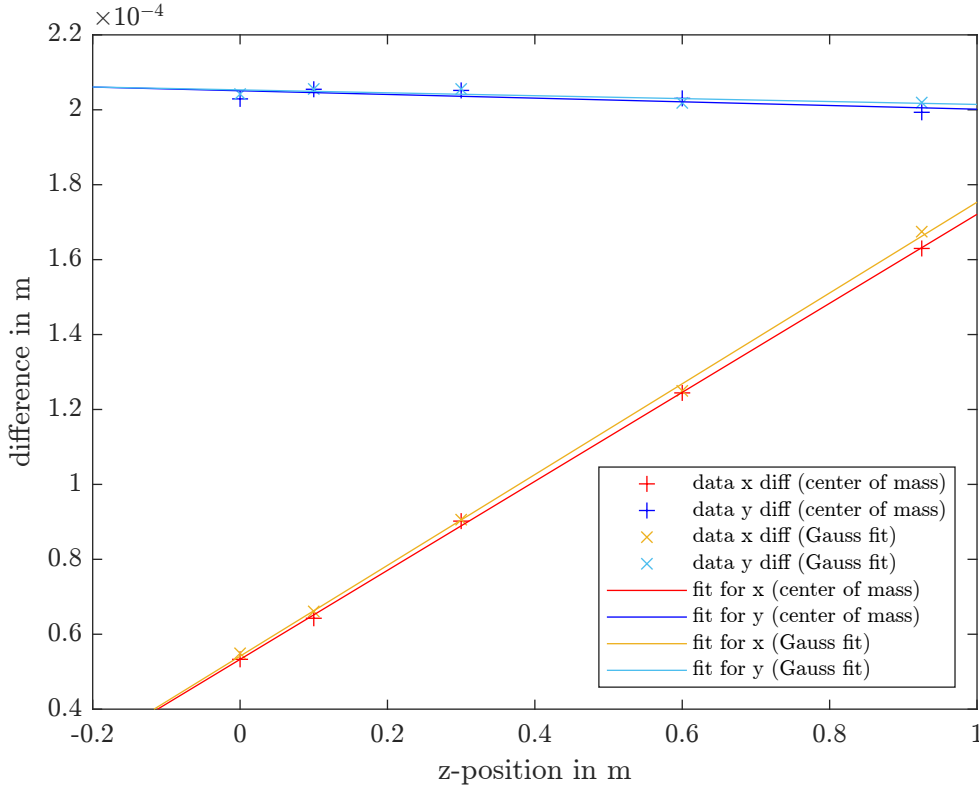


Figure 3.26: Distances between Beams at Different z-Positions. By Köster.

Two methods were used to determine the difference between the beams. Either

3 Experimental setup

the centroids were computed from the beam profiles (in x and y direction), or a Gaussian function was fitted to the intensity distribution and the center extracted. There was no great angular offset in the y-direction. This reflects that the height of the beam was well aligned during the build-up, which certainly was an objective when setting up the interferometer. However, the beams diverge strongly in the x-direction.

Following values were obtained for the angle and shift in both directions:

$$\begin{aligned} \delta\theta_{x,D} &= 120.4 \mu\text{rad}, \\ \delta\theta_{y,D} &= 3.84 \mu\text{rad}, \\ \delta_{x,D} &= 25.91 \mu\text{m}, \\ \delta_{y,D} &= 206.21 \mu\text{m}. \end{aligned} \tag{3.10}$$

Here the index D stands for displacement.

The values for $\delta\theta_x$ from both measuring methods, the phase scan, and the displacement measurement, differ by a factor of -3. This is a rather small difference compared to the difference between $\delta\theta_{y,P,1}$ and $\delta\theta_{y,D}$, which alter by a factor of approximately 40. The great difference in the angle in the y-direction could be caused by the interference fringes observed in the reflected beam (cf. Figure 3.22), which manipulate the interference pattern. In future versions of this experiment, this will be dealt with by setting up a new polarizing beam splitter without these issues, as explained in section 3.3.2.

Nevertheless, both of these deviations are not acceptable. If an error interval estimation was done including intensity fluctuations (around $\pm 50\text{nW}$ observed experimentally), camera inaccuracies (dirty lense, background noise, camera digitization, camera non-linearity, and baseline drift) and considering an error of α , η and ϕ_0 an inaccuracy for $\delta x/\delta y$ of maximally $\approx 30\mu\text{m}$ can be derived. This is surely not great enough to include the previously obtained fit parameters.

After trying different versions of the fit's code and altering fit parameters, an essential observation was made: The fit strongly depends on the value of $\tan(\alpha) = \frac{\sin(\alpha)}{\cos(\alpha)}$, the ratio between both beam's intensities (cf. Equation (2.25)).

The following Figure 3.27 visualizes the real part of the weak value $(\mathbf{P}_A)_w$ of the beam shift (cf. Equation (2.38)) which is dependent on $\tan(\alpha)$. The dotted lines represent the centroids of each separate beam. The position stays constant throughout the phase scan if one measures each beam individually by blocking the other beam because there is no interference. When the beams overlap, the centroid lies between both beams at approximately 0.5mm and phase $\phi = 0$. However, when scanning through the phase, the weak interaction enables an am-

plification of the displacement which is dependent on the weak value (cf. Equation (2.32)). The weak value again includes $\tan(\alpha)$.

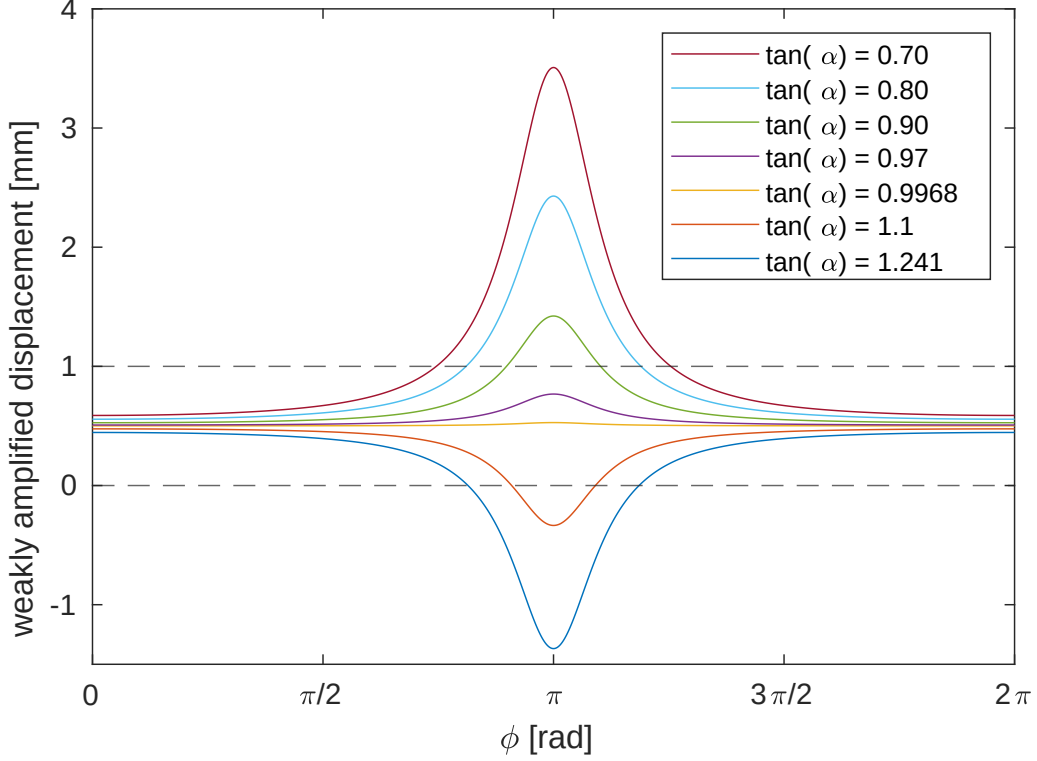


Figure 3.27: Displacement in Dependence of $\tan(\alpha)$.

In this experiment $\tan(\alpha)$ was approximately 1.00 ± 0.03 (cf. Equation (3.6)). In Figure 3.27 one can see that the weak amplification at 1 is almost non-existent. For $\tan(\alpha) = 0.99408$ from Equation (3.6) the amplification factor is given by ≈ 1.1 at its maximum – this can also be seen in Figure 3.25. For this $\tan(\alpha) = 0.99408$, which is smaller than one, the centroid’s amplified displacement happens in the positive x-direction. If $\tan(\alpha)$ was even smaller, the amplification would be much greater, as visualized by the different curves. The same applies to $\tan(\alpha)$ greater than 1, but there the weak amplification takes effect in the opposite direction – this is illustrated by the two curves on the bottom of Figure 3.27.

Similar information can be extracted from the following figure. There, however, various amplifications are shown for different values of η . This plot includes both the real and the imaginary part of the weak value, as in Equation (2.38). Remember that η and $\tan(\alpha)$ are connected via the visibility $\mathcal{V} = \eta \sin(2\alpha)$ from Equation (2.26). The parameter η describes the overlap as written in Equation (2.41). If it reaches 1, the weak amplification diverges, as seen in the figure. Yet, in this experiment, η was 0.9431. Looking at Figure 3.28, the best amplification for this

3 Experimental setup

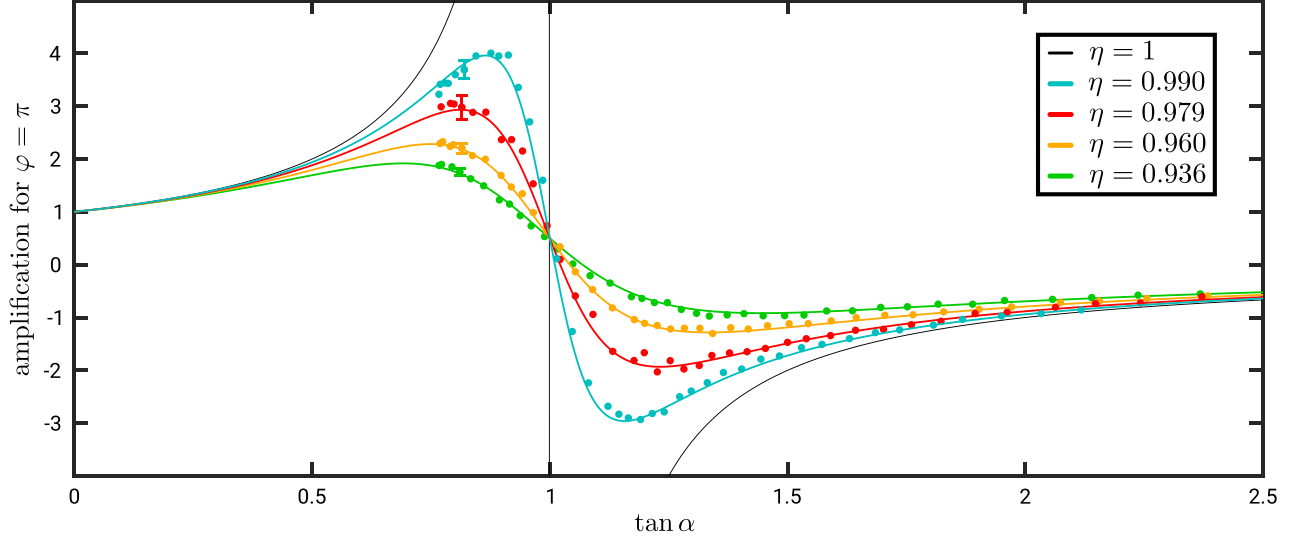


Figure 3.28: Displacement in Dependence of $\tan(\alpha)$ for different η . Taken from [5].

value can be reached through $\tan(\alpha) \approx 0.8$, which yields an amplification factor of roughly 1.7. Nevertheless, the η -value can be increased because it is dependent on α through the visibility (cf. Equation (2.26)). By choosing an even smaller α , the η -value becomes bigger and the amplification greater, as seen in the figure. At $\tan(\alpha) = 1$, however, slight differences in $\tan(\alpha)$ cause the relatively greatest shifts in the amplification because the slope is the steepest at that point. Additionally, this point is an inflection point; for $\tan(\alpha) < 1$ the amplification effect displaces the center of mass in the positive direction, while for $\tan(\alpha) > 1$ the amplification is negative. Altogether, this is why the fit (see Figure 3.25) is highly prone to errors.

Therefore another fit of the same data was done, although this time $\tan(\alpha)$ was not fixed but fitted freely by the software. The fit is illustrated in Figure 3.29. The parameters were computed as

$$\begin{aligned}
 \delta\theta_{x,P,2} &= 99.18 \mu\text{rad}, \\
 \delta\theta_{y,P,2} &= 17.94 \mu\text{rad}, \\
 \delta_{x,P,2} &= 14.44 \mu\text{m}, \\
 \delta_{y,P,2} &= 213.4 \mu\text{m}, \\
 z_{R,x,2} &= 1697 \text{ mm}, \\
 z_{R,y,2} &= 1796 \text{ mm}, \\
 \tan(\alpha)_{x,P,2} &= 1.241, \\
 \tan(\alpha)_{y,P,2} &= 0.9614.
 \end{aligned} \tag{3.11}$$

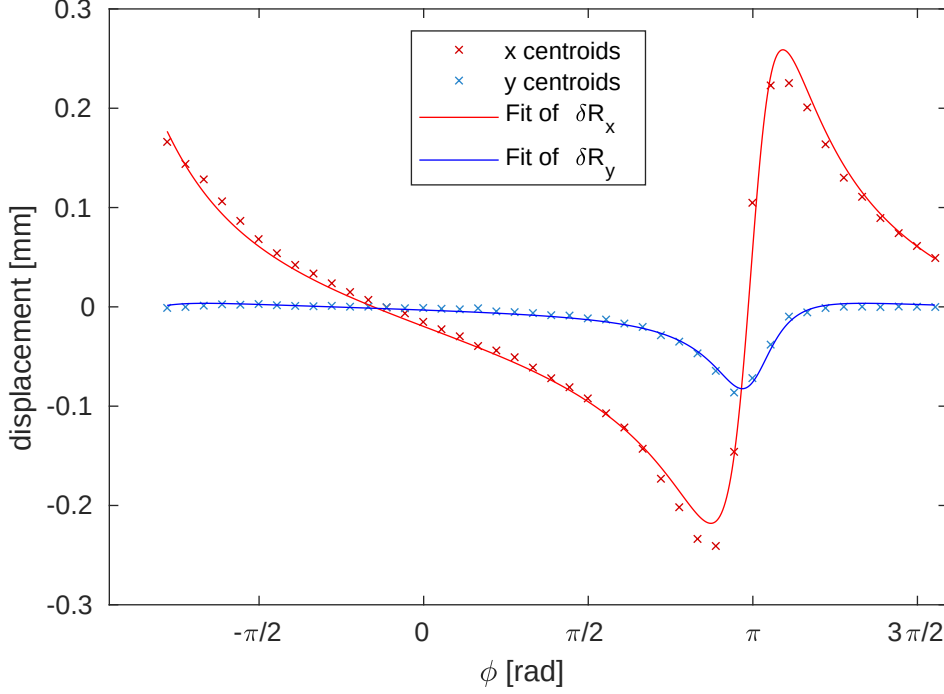


Figure 3.29: Centroids Dependent on ϕ with $\tan(\alpha)$ as Free Fit Parameter.

Most of these values are in harmony with the ones measured at a distance (see 3.10). All deflection parameters, but one, are in the same order of magnitude and differ slightly by factors between 1 and 1.8, which are acceptable error intervals. Still, $\delta\theta_y$ alters by a factor of ≈ 4 . This can be explained through the inaccuracy of $\delta\theta_{y,D}$ because it lies below the measurable resolution.

Furthermore, the Rayleigh length of this fit also agrees with the data from the Gauss fit (cf. section 3.3.3) – this comparison was similarly presented for the Rayleigh lengths of the first fit (cf. Figure 3.25).

Nevertheless, the values for the tangents are not equal, which cannot be the case because the x and y measurements were done simultaneously. In addition, the values for the tangents of α are outside the error interval of $\tan(\alpha)$ from Equation (3.8), which was computed through the observed intensity fluctuations of the laser.

3 Experimental setup

Let us now compare the values obtained for $\tan(\alpha)_{x/y,P,2}$ through the phase scan (see 3.4) to the fitted values for $\tan(\alpha)_{x/y,D}$ of the measurement at a distance. Therefore, the deflection parameters of the measurement at a distance (cf (3.10)) are taken, and put into the equation for $\delta R_{x/y}$, which was fitted already several times (e.g.; Figure 3.25). The same data of the centroids as for all the other fits is used, and $\tan(\alpha)$ is again a free fit parameter.

The values obtained from this fit are $\tan(\alpha)_{x,D} = 1.119$ and $\tan(\alpha)_{y,D} = 1.042$. These values are quite similar to the tangents from the second fit, $\tan(\alpha)_{x/y,P,2}$. Consequently, tangent values can be found for these deflection parameters, and they both must be greater than 1, which physically means that the weak amplification happens in the negative x/y-direction.

Therefore, the value for $\tan(\alpha)_{y,P,2} = 0.9614$ has to be wrong because the shift takes effect in the positive y-direction. Moreover, this can be explained by the camera utilized in the laboratory - it flips images upside down. The blue peak in both Figures 3.25, 3.29 should point upwards.

Therefore, a last fit should be made, where the y-data is flipped, and the tangent values should match with $\tan(\alpha)_{x/y,P,3} > 1$.

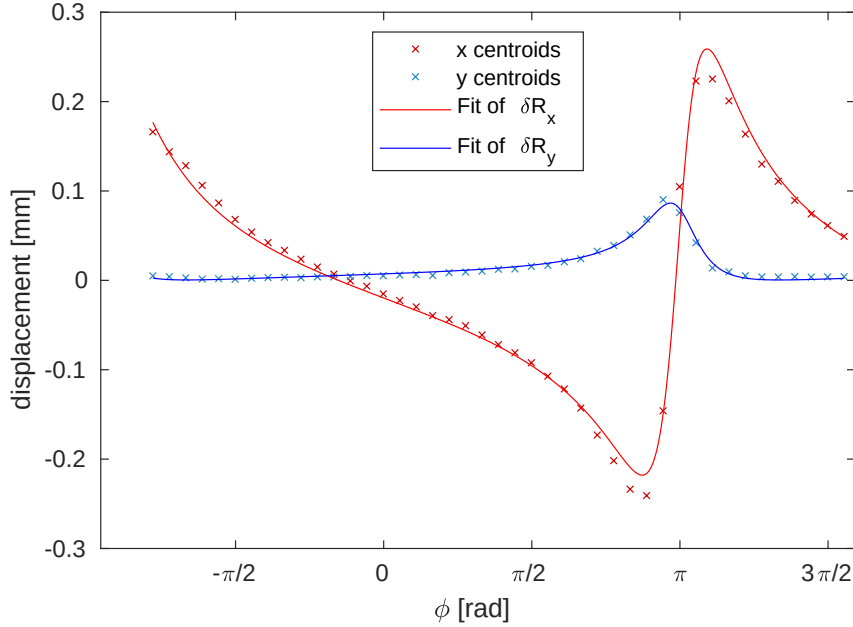


Figure 3.30: Centroids Dependent on ϕ with free $\tan(\alpha)$ and y-Coordinates Mirrored.

The parameters for this fit were realistic. They are given by

$$\begin{aligned}\delta\theta_{x,P,3} &= 99.19 \mu\text{rad}, \\ \delta\theta_{y,P,3} &= 7.3 \mu\text{rad}, \\ \delta_{x,P,3} &= 14.45 \mu\text{m}, \\ \delta_{y,P,3} &= 51.27 \mu\text{m}, \\ z_{R,x,3} &= 1697 \text{ mm}, \\ z_{R,y,3} &= 1734 \text{ mm}, \\ \tan(\alpha)_{x,P,3} &= 1.24, \\ \tan(\alpha)_{y,P,3} &= 1.185.\end{aligned}$$

Fortunately, the tangents have similar values, which is necessary because the intensity ratio does not differ for the x and y coordinates. Furthermore, the values are in an acceptable range of the measurements at a distance. The angle parameters match especially well.

Nonetheless, the ratio of intensities, $\tan(\alpha) = 1.00 \pm 0.03$, chosen in the experiment proves to be the least optimal for actually measuring a strong amplification effect because the amplification is minimal for $\tan(\alpha) = 1$ and slight changes of the tangent alters the weak amplification heavily. Therefore the fit in Figure 3.25 cannot deliver proper values for the angular displacement and spatial shift.

After this first determination of displacement parameters, the mirror should be tilted, and the alignment method should be repeated iteratively, thus, fine-tuning the interferometer to a much higher degree. Potentially, this can reduce the deviations by a factor of 20, as shown in [5]. This will be done in the future by Versmold and Köster.

3.5 Outlook

Since my part in this greater project is done, I will outline the following steps that will be carried out in the future by Versmold and Köster.

Firstly, some optical components like the polarizing beam splitter will be replaced and realigned in the hope of higher efficiency¹². Thenceforward, the interferometer will be aligned iteratively with the alignment method already put into practice, originally from [5]. Again the computed angles can be compared to the actual angles measured at a certain z behind the interferometer.

However, the interferometer cannot be aligned optimally like this because there will be imperfections in the optical components that will not be improved by tilting the mirror. Therefore, a Dove prism has to be inserted into one arm, which deflects the beam inside the interferometer and corrects the imperfections. Then, via another phase scan, the interferometer with the Dove prism (or two Dove prisms for two-dimensional corrections) can be aligned with the same method. Once this is done, the outside beam deflection, d and θ , can be measured using Equation (2.63). The model (cf. section 2.7) in [6] could also be extended into two dimensions, in order to be able to measure $\mathbf{d} = (d_x, d_y)^T$ and $\boldsymbol{\theta} = (\theta_x, \theta_y)^T$. Therefore, a second Dove prism should be placed into the same arm of the interferometer, as one Dove prism can only mirror the beam in one dimension (e.g., about the x -axis). But the second Dove prism could then mirror about the other axis (e.g., y -axis).

Another measurement at two (or more) z -positions can be carried out in order to fit the same Equation (2.63) and determine d and θ .

Furthermore, both methods (phase scan and measurement at various z positions) could be contrasted and weighed.

Another step towards the future of this project is automizing the measurement procedures. This refers to setting up piezoelectric motors that rotate the second half wave plate (to set α) and the third polarizer (to control the projection) and possibly move the mirrors to align the beam.

Moreover, another detector, a position sensing detector instead of the current camera, will be set up that measures more precisely, i.e., $0.75\mu\text{m}$ pixels in place of $4.65\mu\text{m}$ pixels. This position sensing detector can be controlled via a computer, wherefore no measurement must be made manually anymore – this is not only faster but also more precise. Besides, putting a photodiode into place, as mentioned in section 3.3.4 would refine the measurements and enhance the resolution. In the case of the measurement conducted in this paper, a photodiode would have made a great difference since the measurements were very sensitive towards the

¹²The polarizing beam splitter worked with in my setup, BS-40, displayed strange irregularities in the reflected beam which have to be avoided, if one wants to observe a pleasant interference pattern (cf. section 3.3.2).

intensity ratio $\tan(\alpha)$, as shown earlier.

In the long run, a comparison of distinct methods for angle and displacement measurements will be conducted, which will be very helpful in assessing this method among other measurement methods.

3 Experimental setup

4 Conclusions

In this thesis, an alignment method for beam deflections inside an interferometer [5] was presented, and an interferometric measurement of beam deflections outside introduced [6]. First, both methods were described quantum mechanically, as given in [5], [6], and then the focus was put on the experimental implementation of both concepts.

In the laboratory, a Sagnac interferometer was set up. Firstly, the laser beam and optical components were characterized and the interferometric setup comprehensively analyzed. Following, the interferometer was built and adjusted via the alignment method. A phase scan enabled the determination of beam displacement parameters caused by imperfect components inside the interferometer. In this way, both the beam's spatial shift and the angular displacement were computed.

Afterwards, this method was compared to measuring the beam's relative angle and displacement over a large distance. The results demonstrated a great dependence on the intensity ratio between both arms of the interferometer, given by $\tan(\alpha)$, which was thoroughly examined. The final displacement parameters found from the alignment method were consistent with the values obtained at a distance.

A short outlook on the future of this experiment was given, like the setup of the Dove prism to measure beam deflections outside the interferometer. A few improvements were named that were already conceived by Versmold and Köster, who will take over the experiment and make further measurements.

4 Conclusions

Abbreviations

AAV	Aharonov, Albert and Vaidman
BS	beam splitter
CAM	camera
DP	dove prism
FC	fiber coupler
H	horizontal
HWP	half wave plate
M	mirror
P	prism
PBS	polarizing beam splitter
PD	photo diode
POL	polarizer
QWP	quarter wave plate
V	vertical

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Declaration of authorship – Selbstständigkeitserklärung

I hereby declare that I have written this thesis independently and that I have not used any sources or aids other than those stated in the thesis.

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